

Co-Designing Mechanisms and Motion: Towards Safe and Deterministic Human–Robot Interaction

Durgesh Haribhau Salunkhe

Requested host institutions:

1. ICube, Strasbourg: Team RDH
2. LAAS, Toulouse: Team Gepetto



An urgent need!

■ Blocs opératoires : la délégation d'actes infirmiers jusque-là réservés aux Ibode jette le trouble

PAR ARNAUD JANIN - PUBLIÉ LE 24/01/2025

0 RÉACTIONS COMMENTER



Le Quotidien du Médecin, 24 Jan 2025

Parution du décret permettant aux IDE, à titre transitoire et sur autorisation, la pratique des actes exclusifs IBODE



03 Feb, 2025

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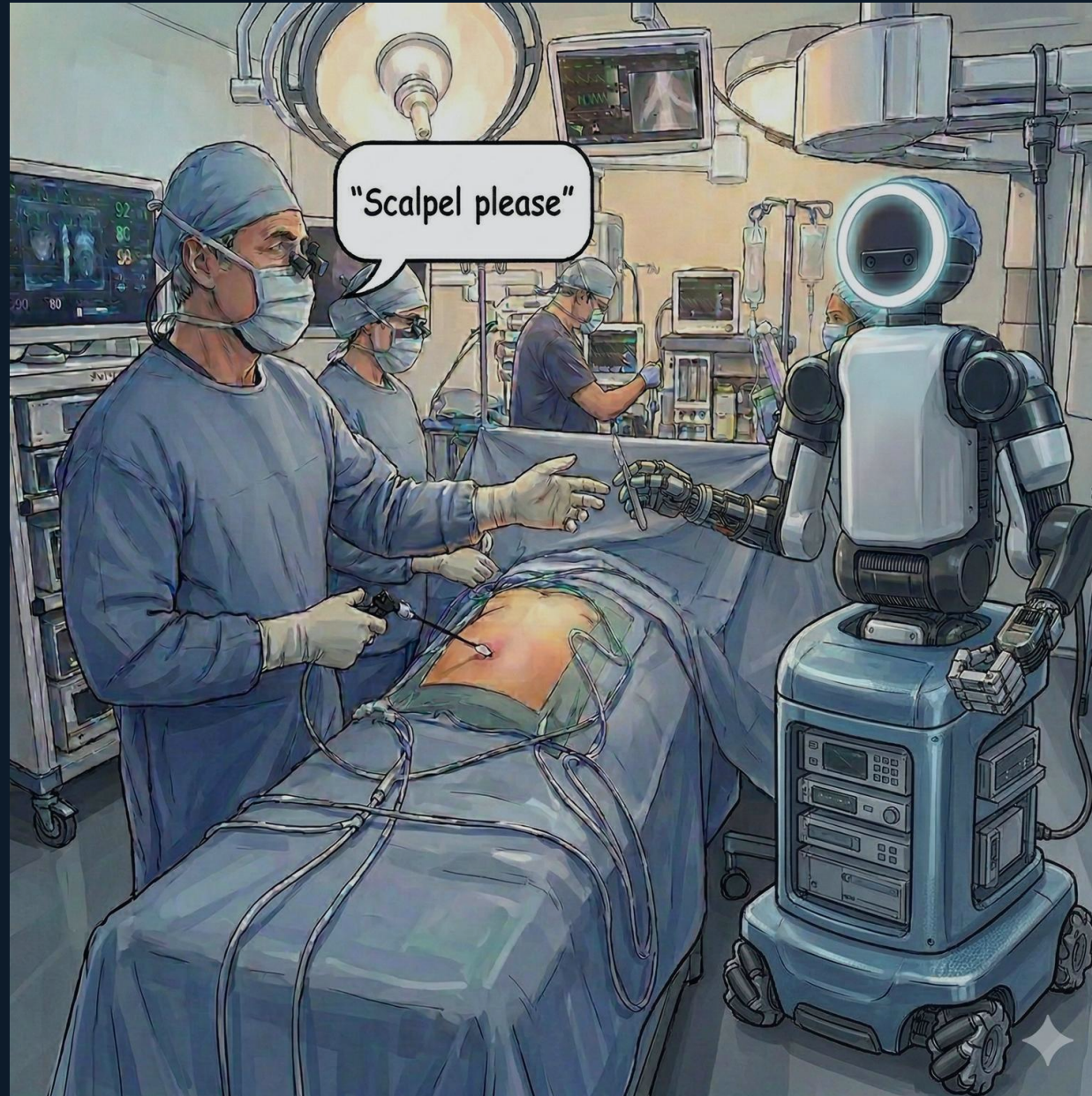
03 Feb, 2025



Versius system (UK) used in Chemnitz Hospital, Germany

Please pass me the scalpel

An example scenario



AI-generated image

The three barriers to ubiquitous robots

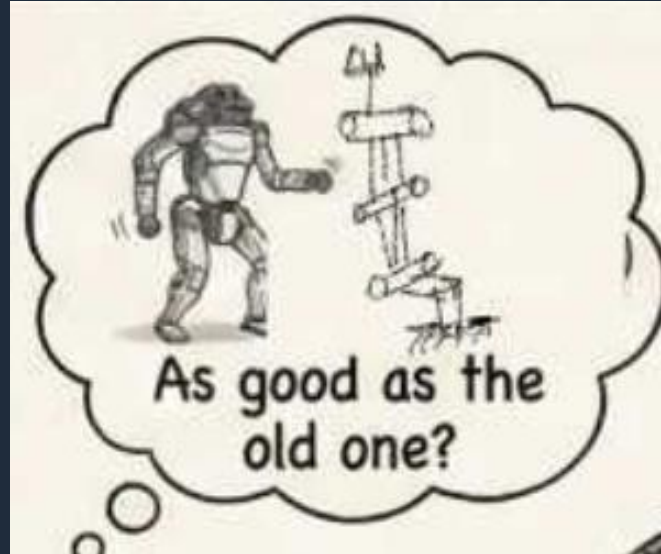
Why a surgeon may hesitate to trust a robot

Reliable Execution



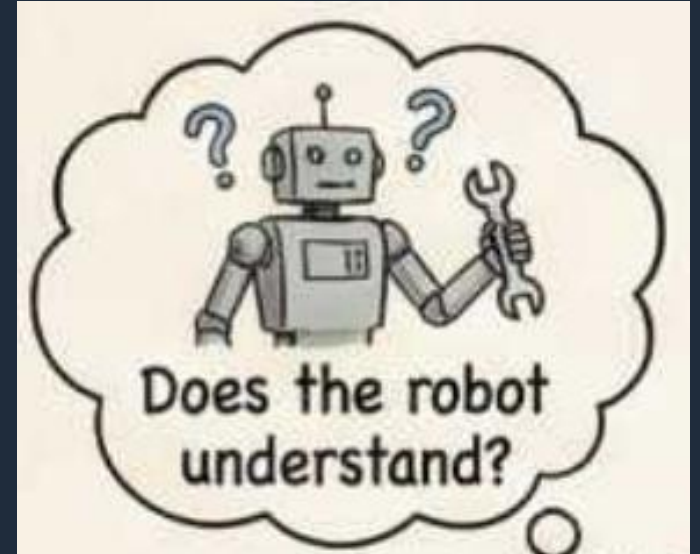
The real world is unstructured: robot execution must be safe and explainable.

Skill Scalability



As new robots are developed, skills need to safely transferred across all generations of robots.

Embodied alignment

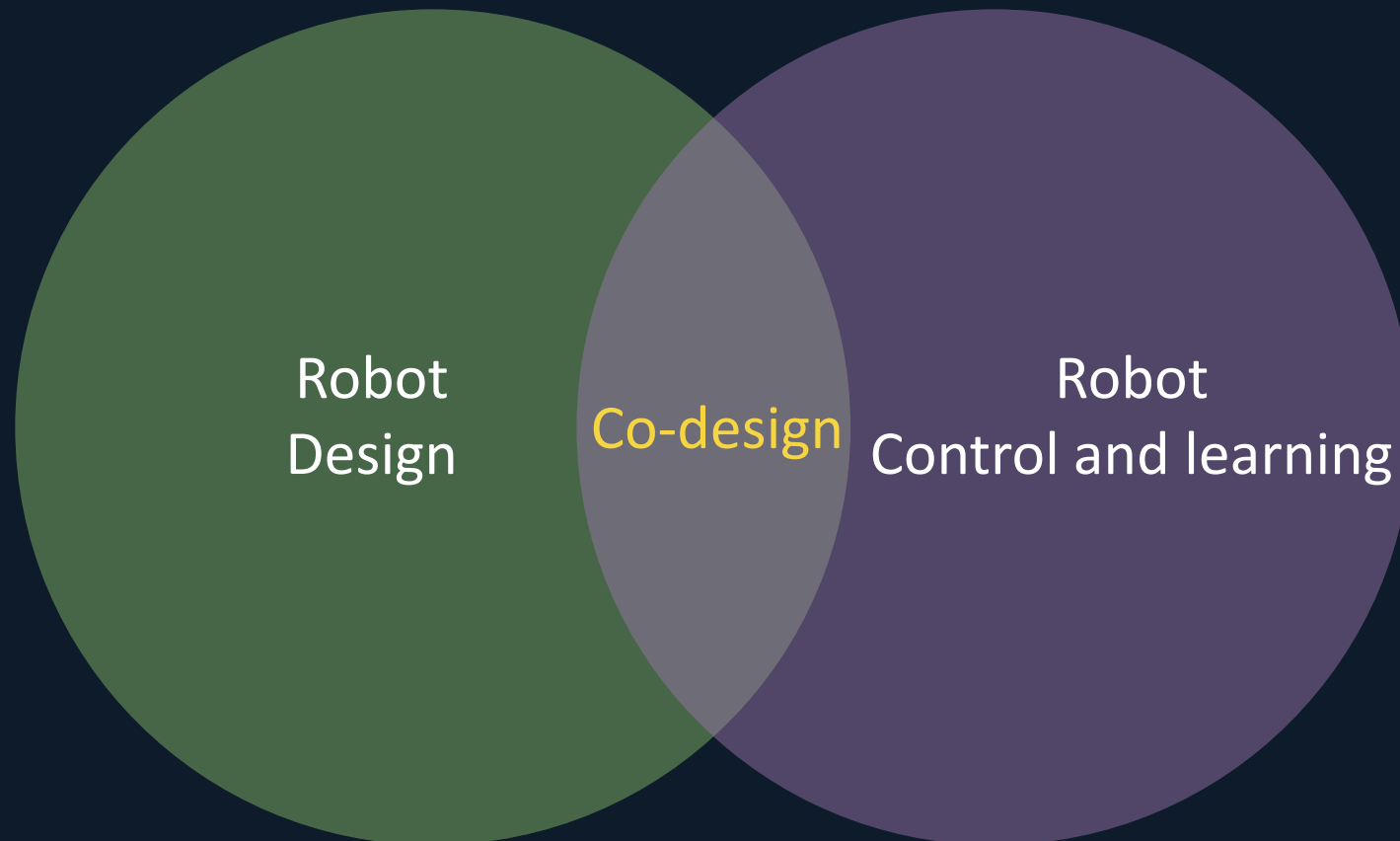


Robot design and actions should be kinematically-grounded while working with LLM and GenAI

Research Approach

***Co-design approach:** to develop robot designs and learning paradigms together, from the ground up, for human-centric tasks.*

(Cheney et al., '18)



Research timeline

Research Theme 1: Task-Oriented Robot Design

1. Design optimization of robots for otological surgery

Journals: MMT'20

Conferences: ICRA'22

Software: ParaOpt (used by LS2N, DFKI)

Research Theme 2: Safe and Reliable Robot Behavior

1. Active force compliance in 6-RSS parallel platform

Journals: RCIM'19

Conferences: AIR'20

Software:

2018-20



Director: Damien Chablat

Collaborations:

CHU Nantes, DFKI Germany

Research timeline

Research Theme 1: Task-Oriented Robot Design

1. Design optimization of robots for otological surgery
2. Kinematic analysis of generic 3R and 6R robots

Journals: MMT'20, MMT'22

Conferences: ICRA'22, ARK'22, ISSAC'22

Software: ParaOpt (used by LS2N, DFKI)

Research Theme 2: Safe and Reliable Robot Behavior

1. Active force compliance in 6-RSS parallel platform
2. Path planning framework for 6R robots

Journals: RCIM'19, IJRR'24

Conferences: AIR'20, ICRA'23, IROS'23

Software:

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2020-23



Director: Philippe Wenger, Damien Chablat

Collaborations with:

Sorbonne, Linz, Innsbruck

Research timeline

Research Theme 1: Task-Oriented Robot Design

1. Design optimization of robots for otological surgery
2. Kinematic analysis of generic 3R and 6R robots
3. Novel designs for Human-Robot Interaction

Journals: MMT'20, MMT'22, RA-L'25

Conferences: ICRA'22, ARK'22, ISSAC'22, CK'25, IMA'25

Software: ParaOpt (used by LS2N, DFKI)

Research Theme 2: Safe and Reliable Robot Behavior

1. Active force compliance in 6-RSS parallel platform
2. Path planning framework for 6R robots
3. Kinematic Intelligence for cross-robot skill transfer

Journals: RCIM'19, IJRR'24, Science Robotics '26

Conferences: AIR'20, ICRA'23, IROS'23

Software: Skill Transfer (April '26)

2018-20



Director: Damien Chablat

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Director: Philippe Wenger, Damien Chablat

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2024-Current



Supervisor: Aude Billard

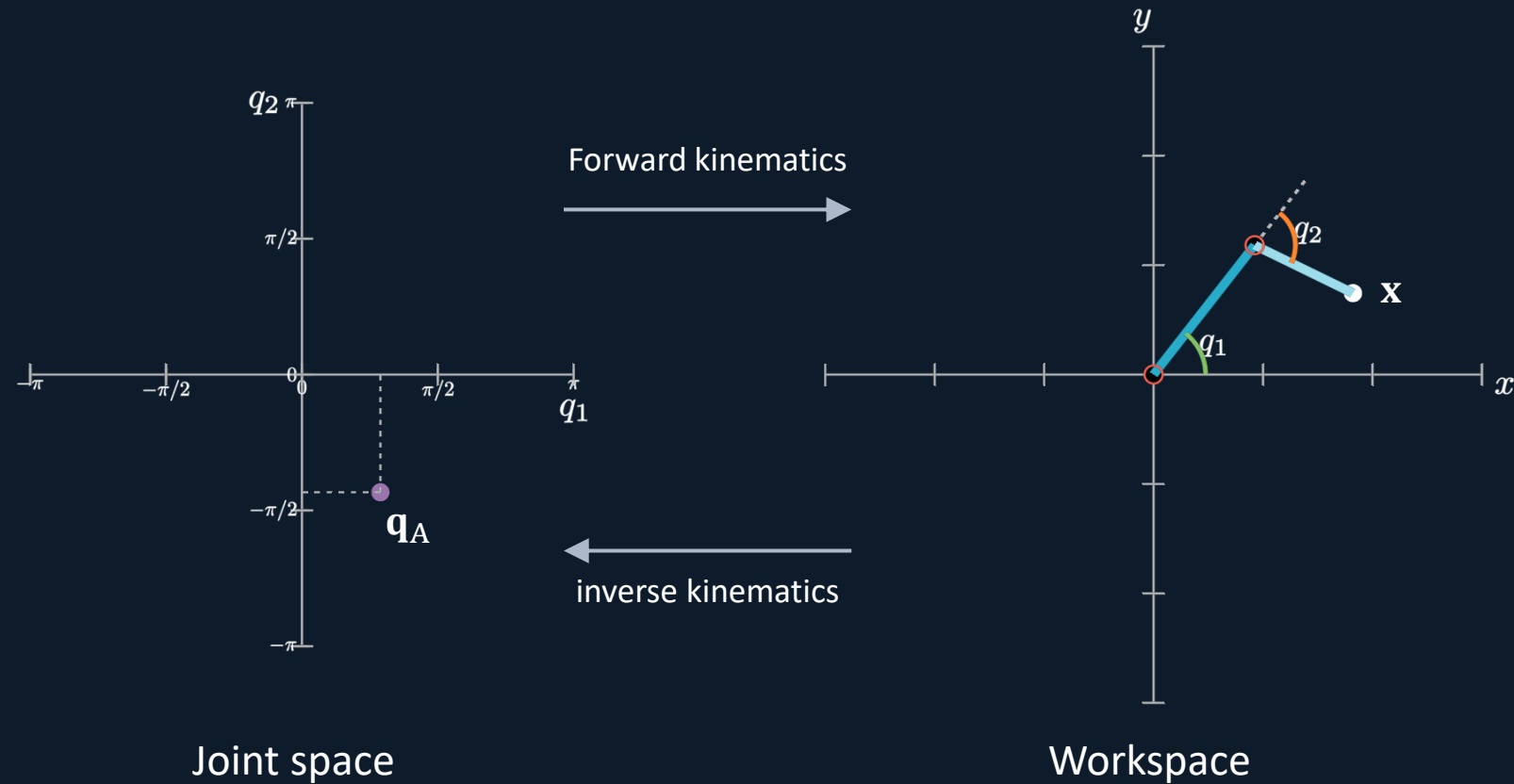
Collaborations with:

IIT, India, IRI, Spain

Research activity

Doctoral work

nR robot: A robot with 'n' revolute joints. (I will use 2R, 3R, 6R and 7R)

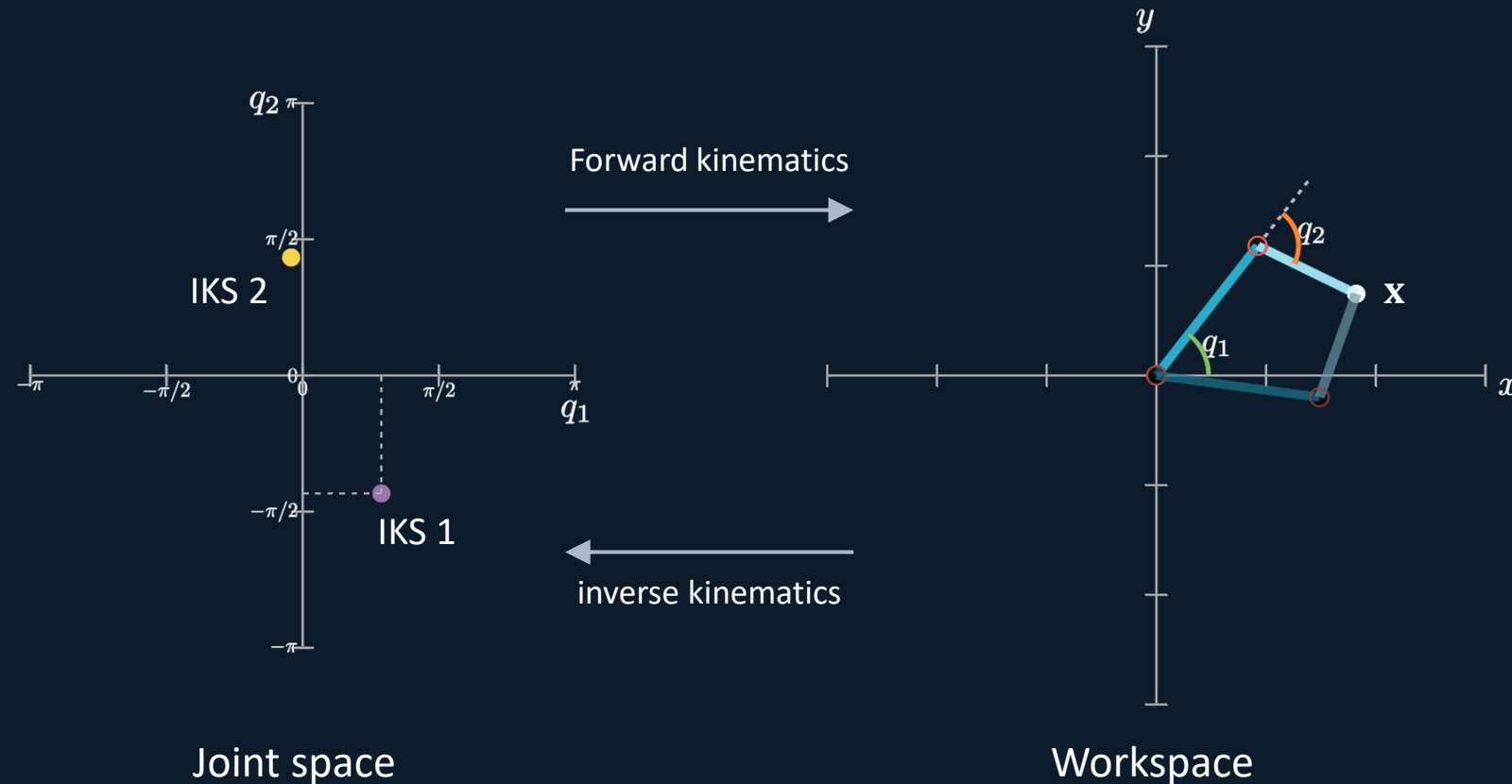


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Inverse kinematic solutions(IKS): different numbers of ways a robot can reach the same target



Research activity

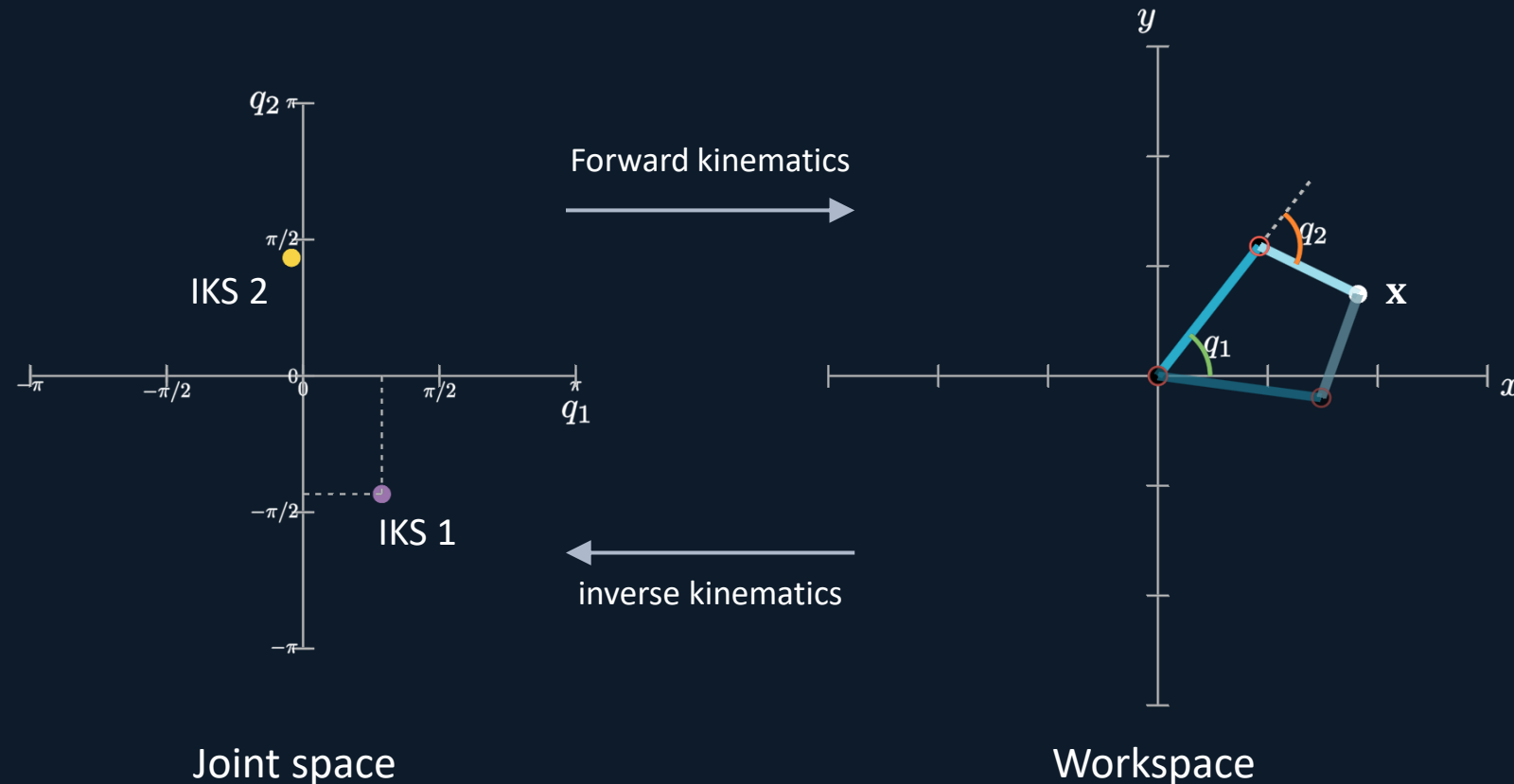
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Jacobian: maps the joint space velocities to workspace velocities

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$



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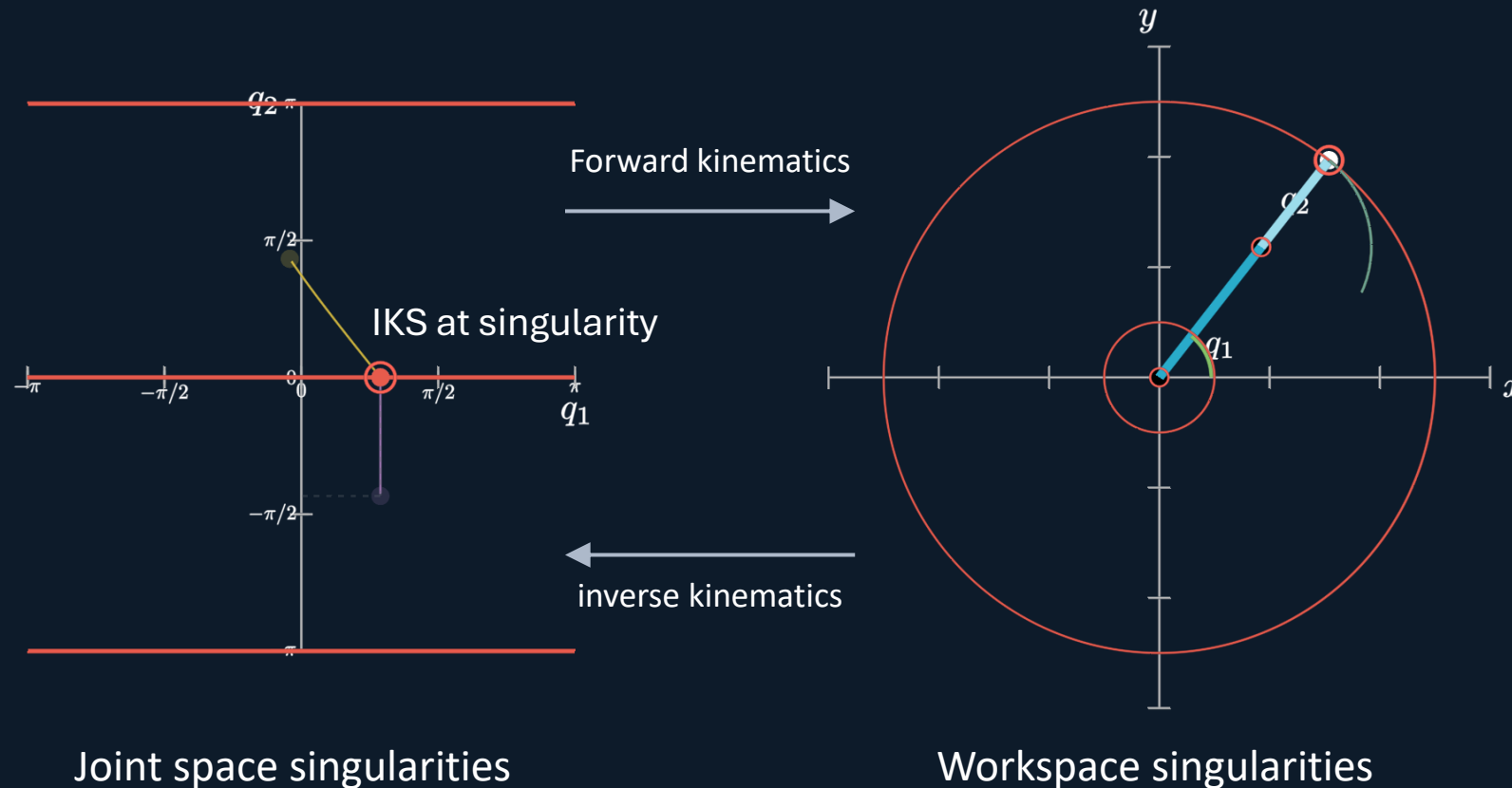
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Singularity: Configurations where Jacobian is not full rank

$$\det \mathbf{J}(\mathbf{q}) = 0$$



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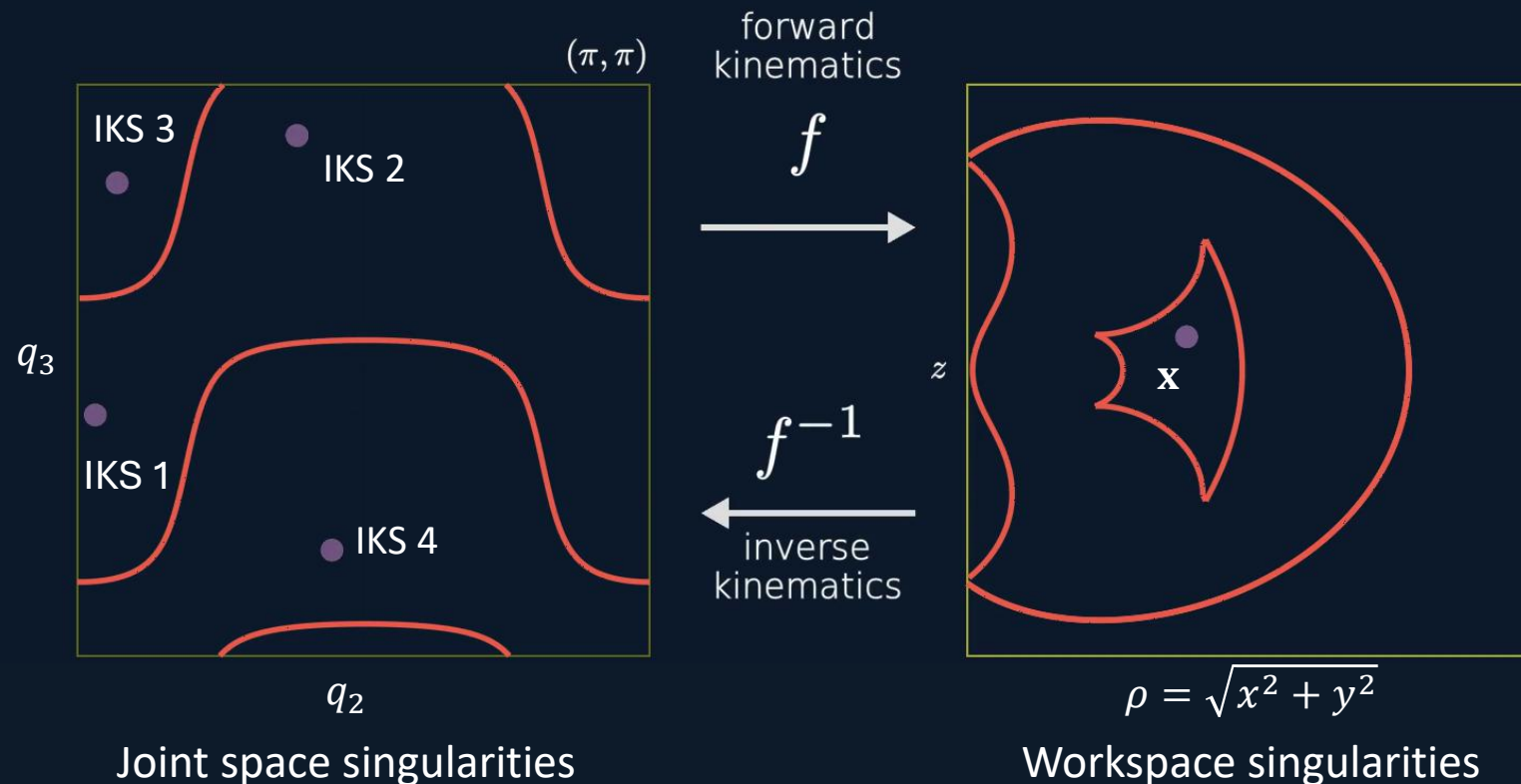
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Cuspidal Robots: Robots that change inverse kinematic solutions without crossing a singularity



Research activity

Doctoral work

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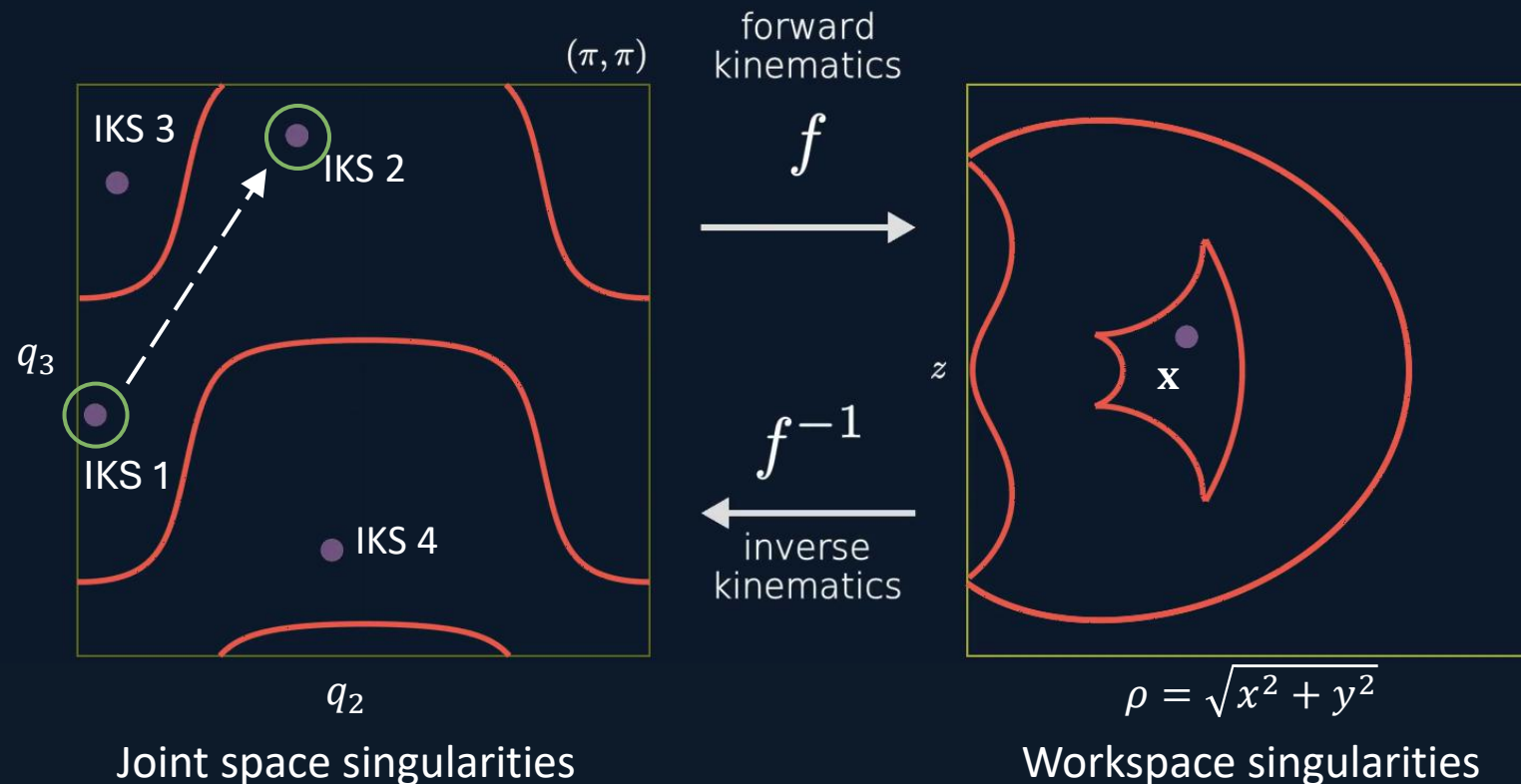
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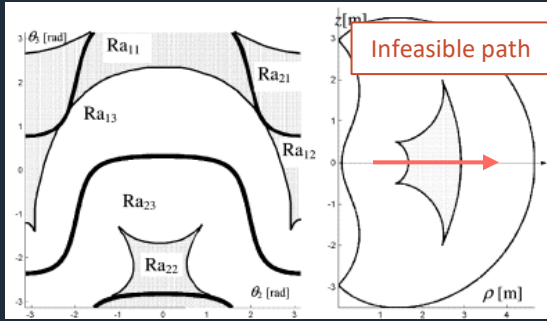


Research activity

Doctoral work

PREVIOUS WORK

- Trajectory planning issues in 3R cuspidal robots



(Wenger '04)

- Cuspidality analysis of 6R robots



UR5 is a
noncuspidal robot

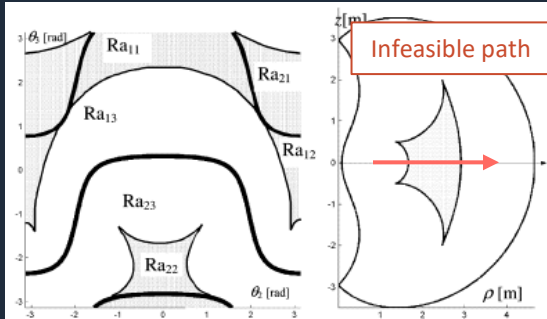
(Capco et al. '20)

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1. How to determine if a given 6R robot is cuspidal or not?

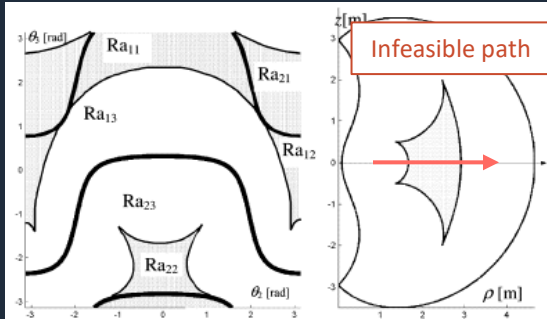
2. What are the implications on path planning for a 6R cuspidal robot?

Research activity

Doctoral work

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(Capco et al. '20)



Certified IK + Certified connectivity
=
Deciding cuspidality

Result 1:

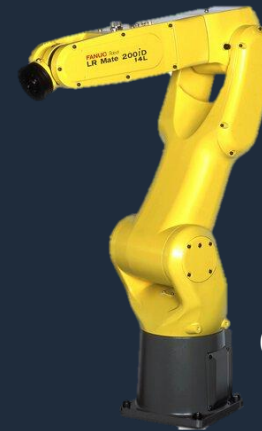
Even a small deviation from conventional design
almost always leads to cuspidal robots.

(ISSAC'22, IJRR'24)

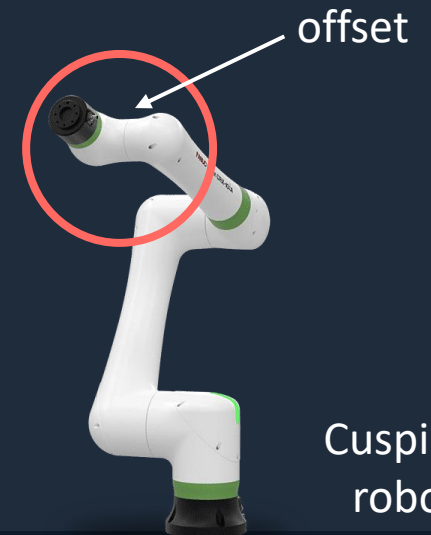


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2. What are the implications on path planning
for a 6R cuspidal robot?



Conventional
robot



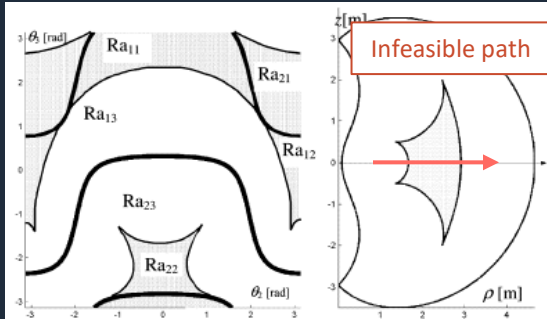
Cuspidal
robot

Research activity

Doctoral work

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(Wenger '04)

- Cuspidality analysis of 6R robots



UR5 is a
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(Capco et al. '20)



Perform workspace analysis and explore different path properties for cuspidal robots.

Result 1:

Even a small deviation from conventional design almost always leads to cuspidal robots.

(ISSAC'22, IJRR'24)

Result 2:

A framework for safe and reliable path planning of commercial 6R robots.

(ICRA'23, IJRR'24)



1. How to determine if a given 6R robot is cuspidal or not?

2. What are the implications on path planning for a 6R cuspidal robot?

Research activity

Postdoctoral work



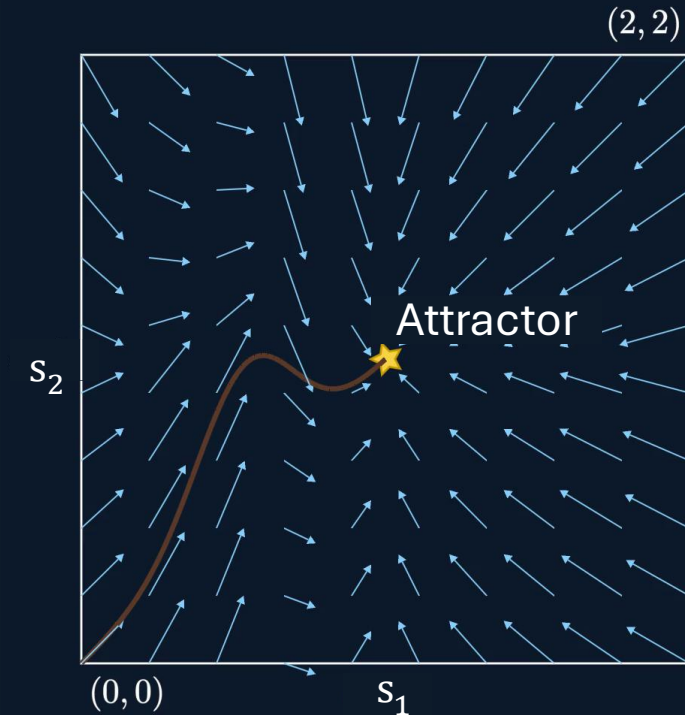
1. Explainable transfer of a user's demonstrated behaviour across different robots.



Research activity

Postdoctoral work

Control policy: maps the robot's current state to its next action or velocity command $\dot{s} = f(s)$



A control policy can be visualized as a vector field in the parameter space



1. Explainable transfer of a user's demonstrated behaviour across different robots.

Research activity

Postdoctoral work

PREVIOUS WORK

Morphology-aware transfer

1. Infer the robot morphology (Trabucco et al. '22)
2. Hint the learning model about robot structure with biases (Sferrazza et al., '24)
3. Train a single policy that handles robot variations (Mu et al, '26, Yang et al., '26)



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Research activity

Postdoctoral work

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1. Explainable transfer of a user's demonstrated behaviour across different robots.



Embedding robot structure into learning for safe and deterministic motion.

Stable robot-agnostic policy
+
Deterministic robot-specific execution
=
Explainable cross-robot skill transfer

1. Introduced a novel paradigm for an explainable cross-robot skill-transfer framework ([Science Robotics '26](#))
2. Collaboration with IRI Spain on kinematic analysis using Conformal Geometric Algebra ([IMA '25 x2](#))

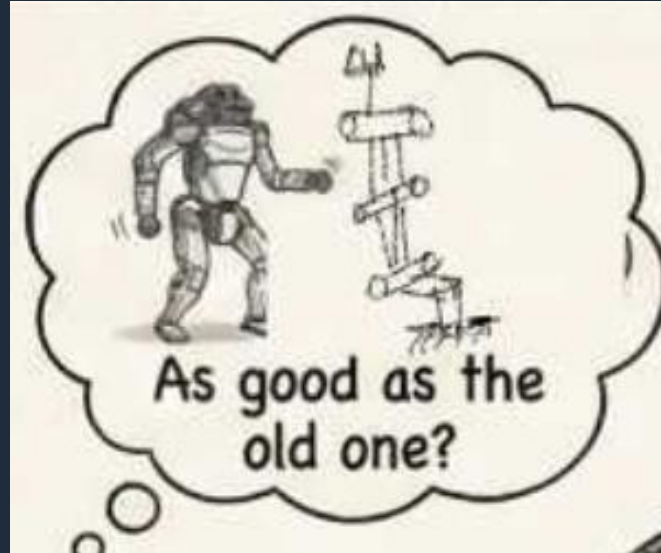
Research plan

Scientific focus

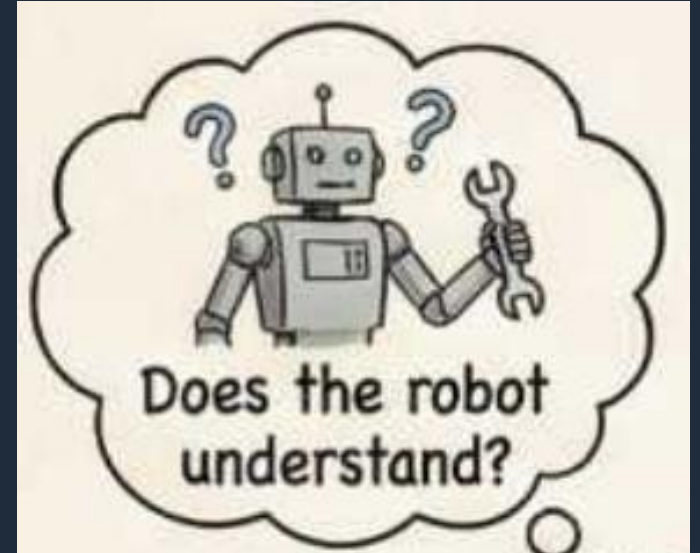
Reliable Execution



Skill Scalability



Embodied alignment



Short-term

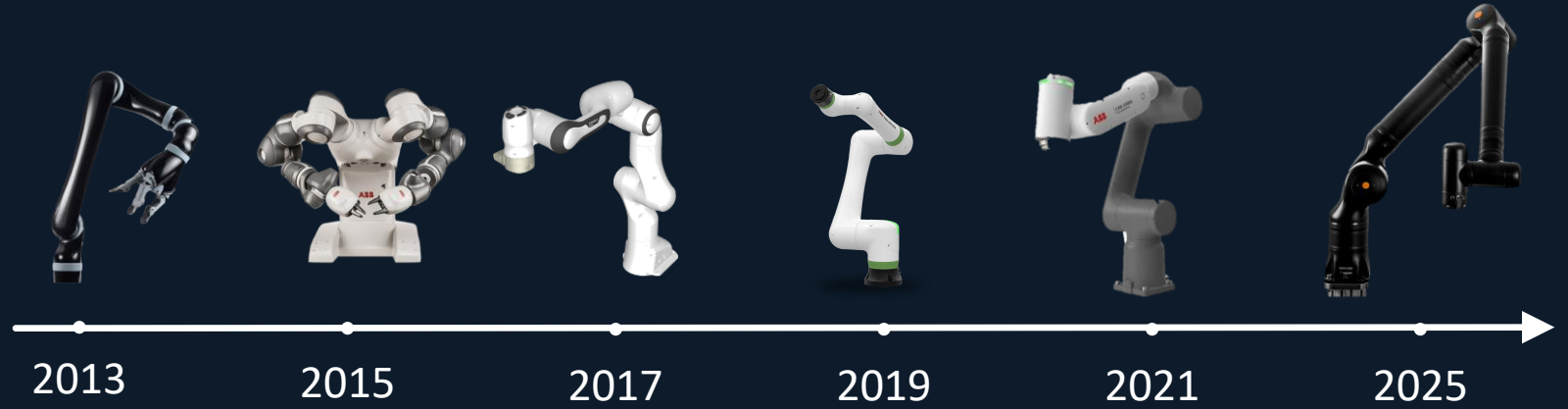
Mid-term

Long-term

Phase 1

Short-term scientific focus

Reliable Execution



1. Many modern commercial robots (6R and 7R) are cuspidal.
2. Almost all of them are used for collaborative purposes.

Analysis of cuspidal robots is both urgent and essential.

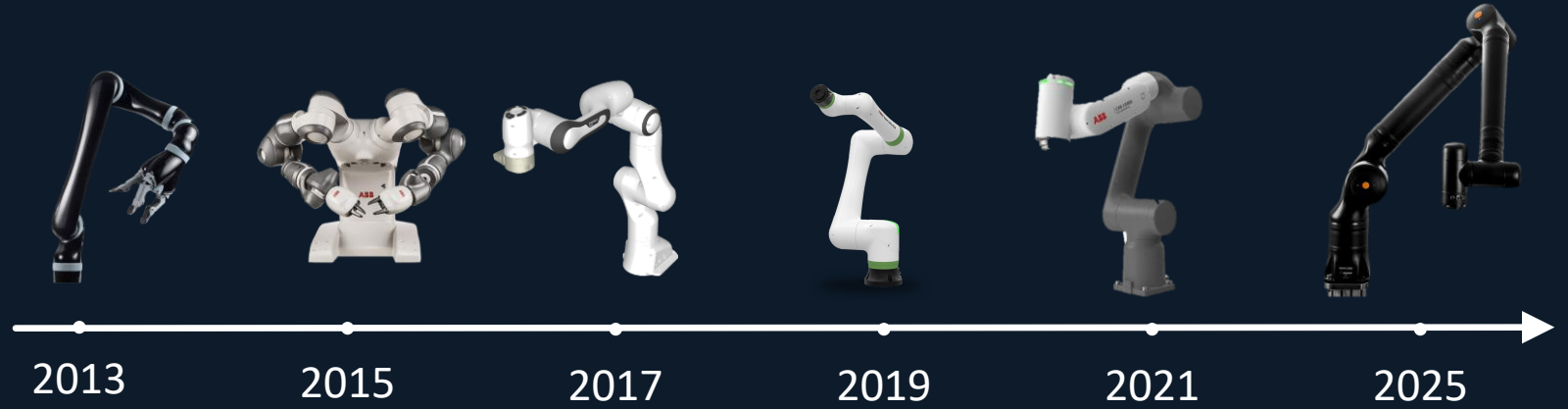
Phase 1

Short-term scientific focus

Reliable Execution



1. **Design:** Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.



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Phase 1

Short-term scientific focus

Reliable Execution



1. **Design:** Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.



Co-design approach:

1. **Design:** Perform global kinematic analysis for cuspidal robots.

1. Use accelerated processing and efficient algorithms for workspace analysis

Efficient
IK solvers

+

Reduced-dimension
analysis

+

Certified cell
decomposition

=

Global kinematics
of cuspidal robots

Conformal
Geometric
Algebra

Redundant angle
parameterization

Cylindrical
Algebraic
Decomposition

Phase 1

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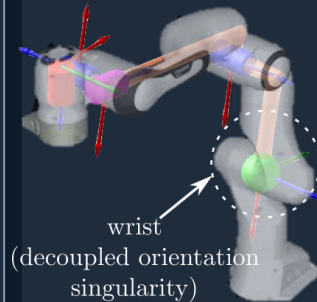
Global kinematics
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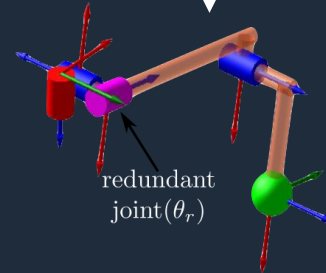
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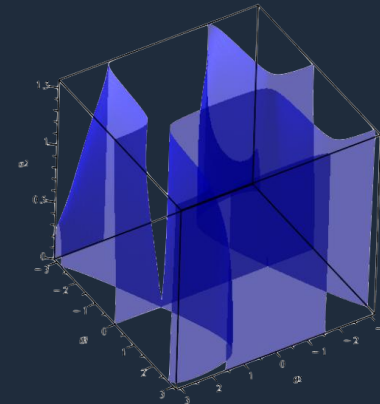
example
↓



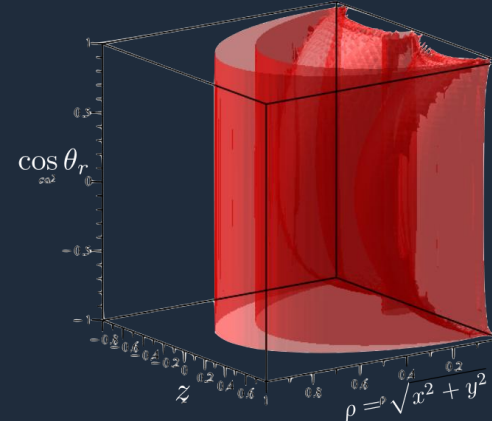
7R robot



4R + 3R robot



Stacked joint space
singularities



Stacked workspace
singularities

Phase 1

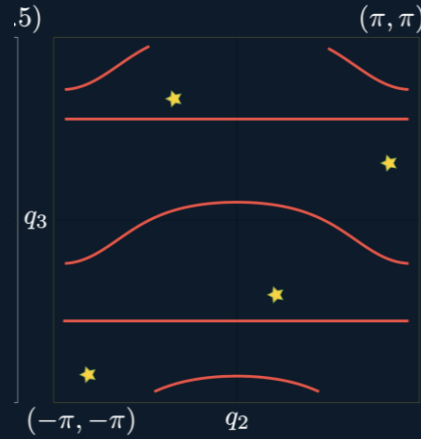
Short-term scientific focus

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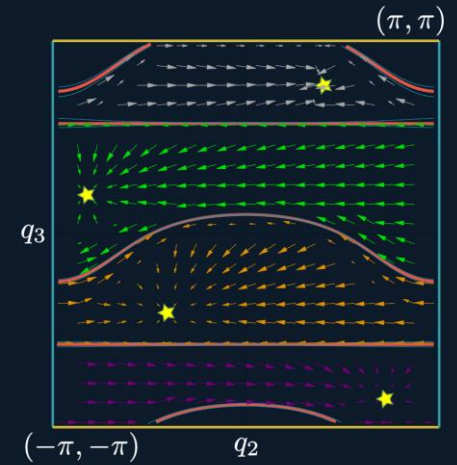
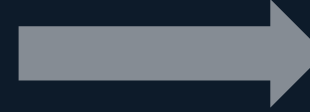
1. **Design:** Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.
2. **Control:** No unique isolated control policies for cuspidal robots.

Noncuspidal robot



Joint space
'star' is IKS of attractor

One IKS $\rightarrow$ One control policy



Joint space policy
Individual policy in each region

Phase 1

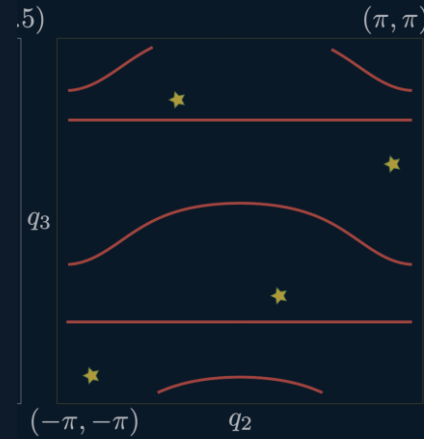
Short-term scientific focus

Reliable Execution



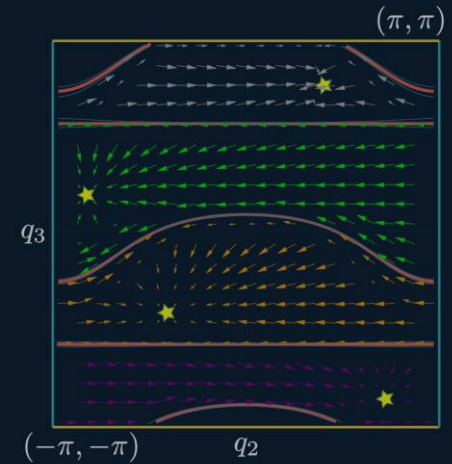
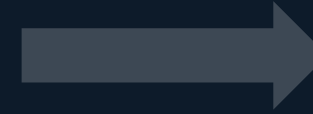
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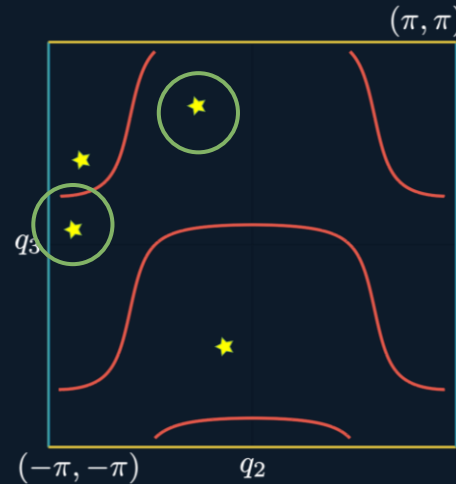
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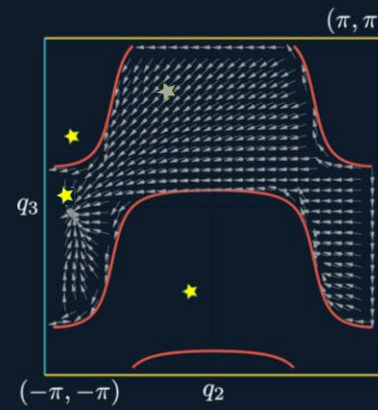
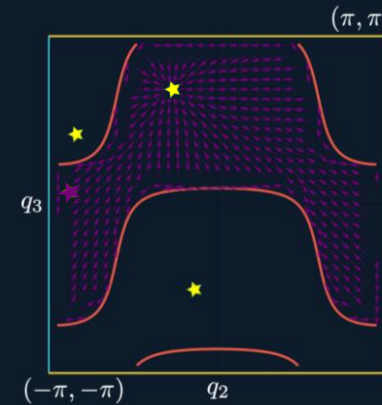


Joint space policy
Individual policy in each region

Cuspidal robot



??



Which control
policy to use?

Phase 1

Short-term scientific focus

Reliable Execution



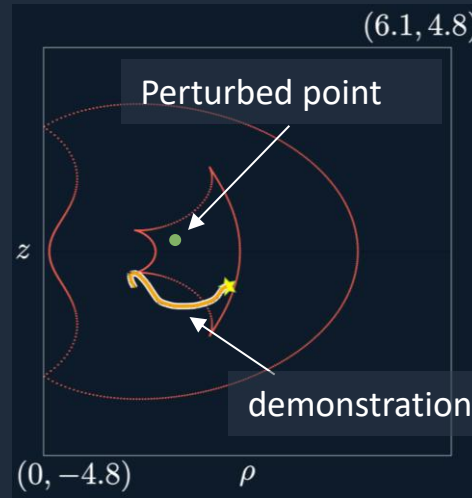
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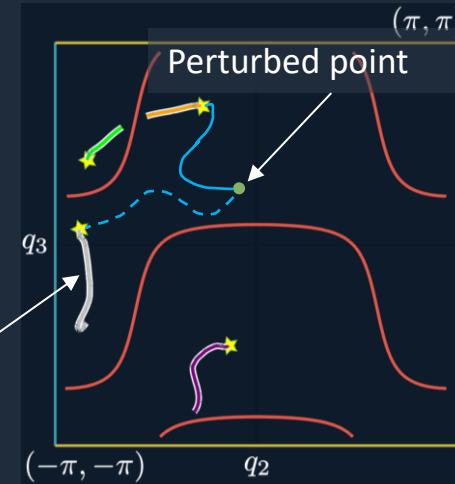
Co-design approach:

1. **Design:** Perform global kinematic analysis for cuspidal robots.
2. **Control:** Develop metrics to identify the suitable control policy

2. Study a metric to enquire proximity to user intent



workspace



joint space

Which path do I choose?

Leverage **Kinematic Intelligence**

Phase 1

Short-term scientific focus

Reliable Execution



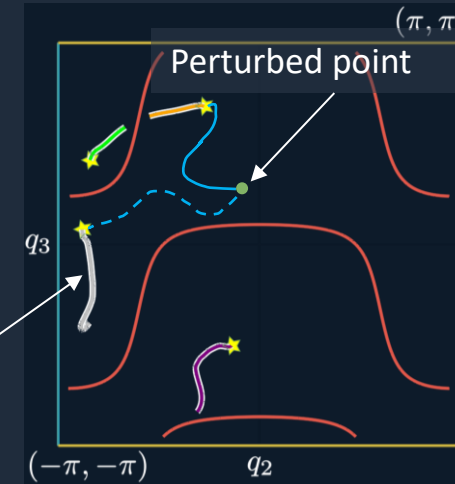
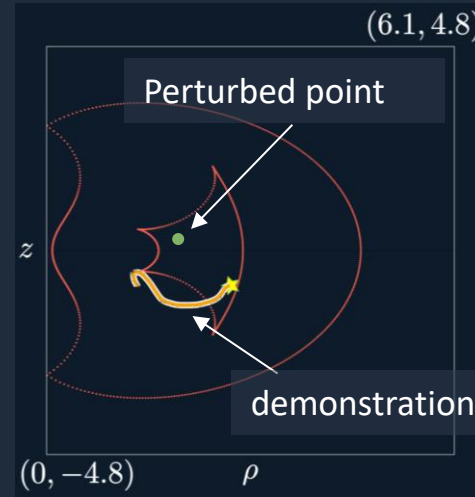
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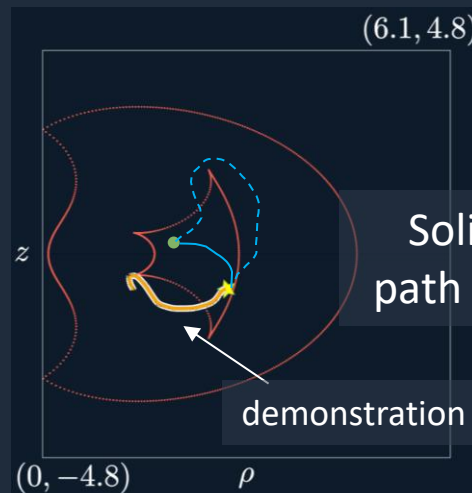
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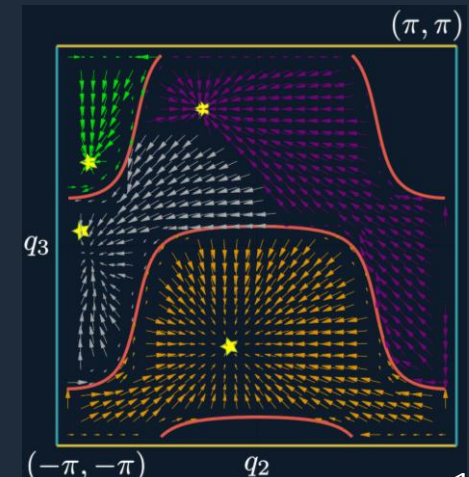


Which path do I choose?

Use fast all-at-once forward mapping and measure the deviation from the intended behaviour



Expected final control policy



Phase 2

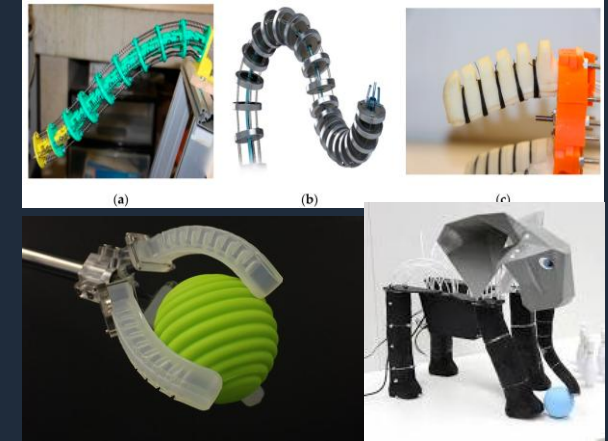
Mid-term scientific focus

Skill Scalability

Rigid links



Soft material



Increasing flexibility

Increasing precision

Phase 2

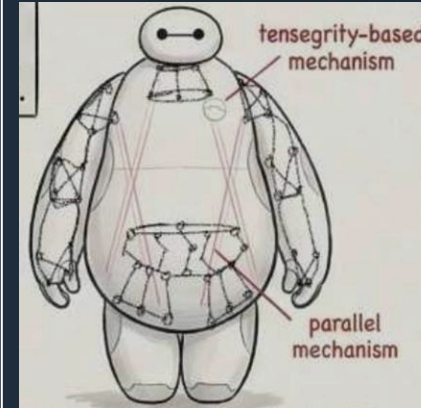
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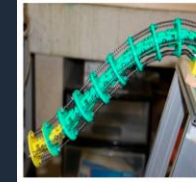
Rigid links



Robots for HRI



Soft material



Increasing flexibility

Increasing precision

Properties desired in human-robot interaction (Haddadin and Croft, '16):

1. Mechanical impedance
2. Lightweight designs
3. Transparency (natural motion)

Phase 2

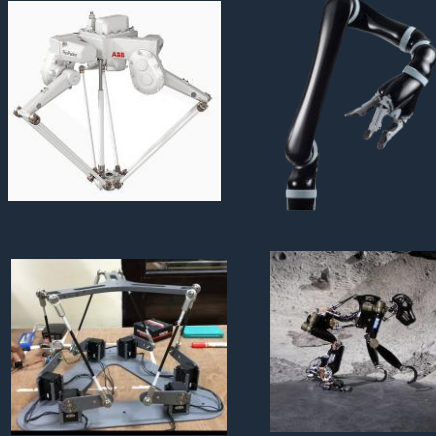
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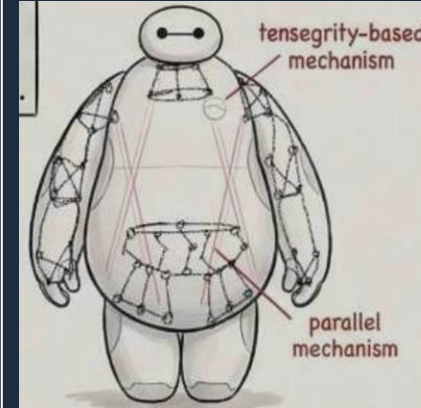


1. **Design:** Robots are either precise and rigid or soft and compliant.

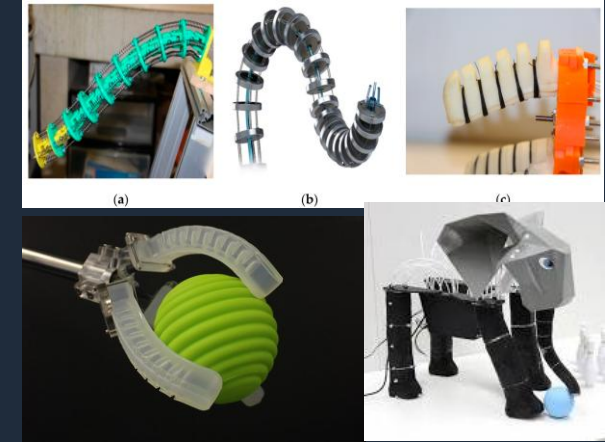
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Robots for HRI



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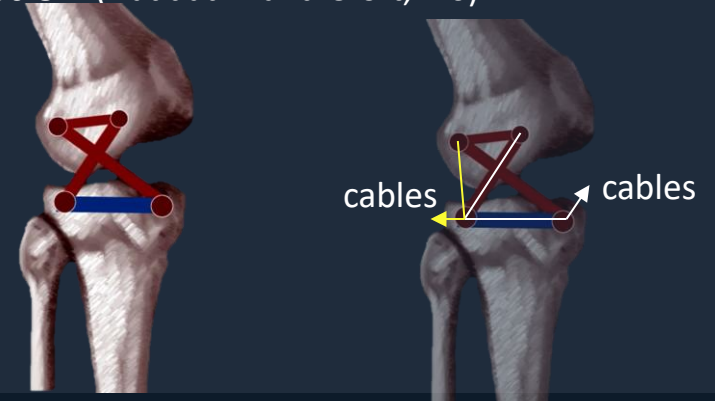


Increasing flexibility

Increasing precision

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1. Mechanical impedance
2. Lightweight designs
3. Transparency (natural motion)



Phase 2

Mid-term scientific focus

Skill Scalability



1. **Design:** Robots are either precise and rigid or soft and compliant.



Co-design approach:

1. **Design:** Rigid–soft trade-off through novel joints and mechanisms

1. Rigid–soft trade-off through novel joints and mechanisms

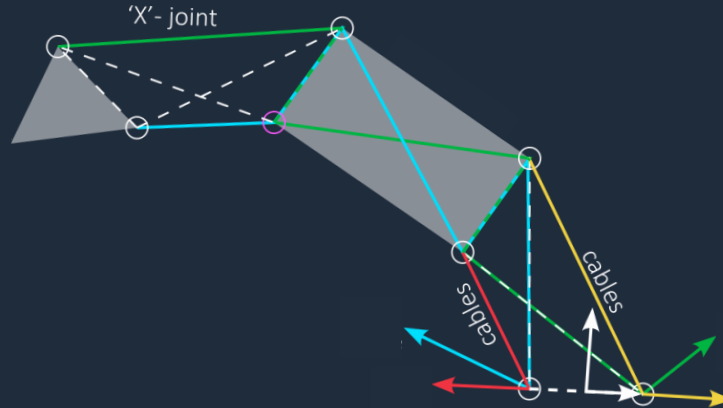
Mechanisms that mimic human joints

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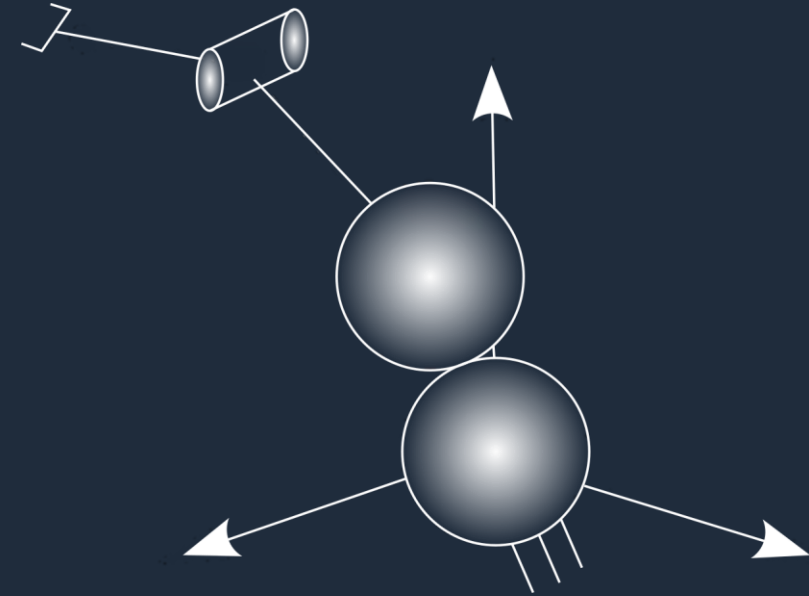
Cable-driven actuation

=

Light robots with variable stiffness



Cable-driven 'X'-joints
(Muralidharan et al., '25)



Series-parallel arm with sphere-on-sphere rolling kinematics

Explore hybrid designs and novel actuations to maximize the desirable properties for HRI

Phase 2

Mid-term scientific focus

Skill Scalability



1. **Design:** Robots are either precise and rigid or soft and compliant.

Co-design approach:

1. **Design:** Rigid–soft trade-off through novel joints and mechanisms



1. Rigid–soft trade-off through novel joints and mechanisms

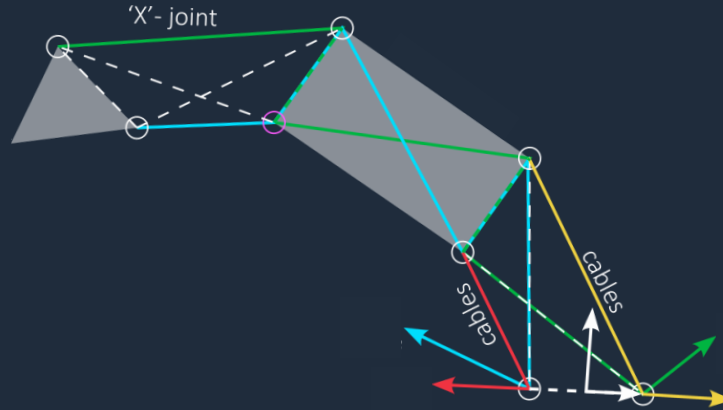
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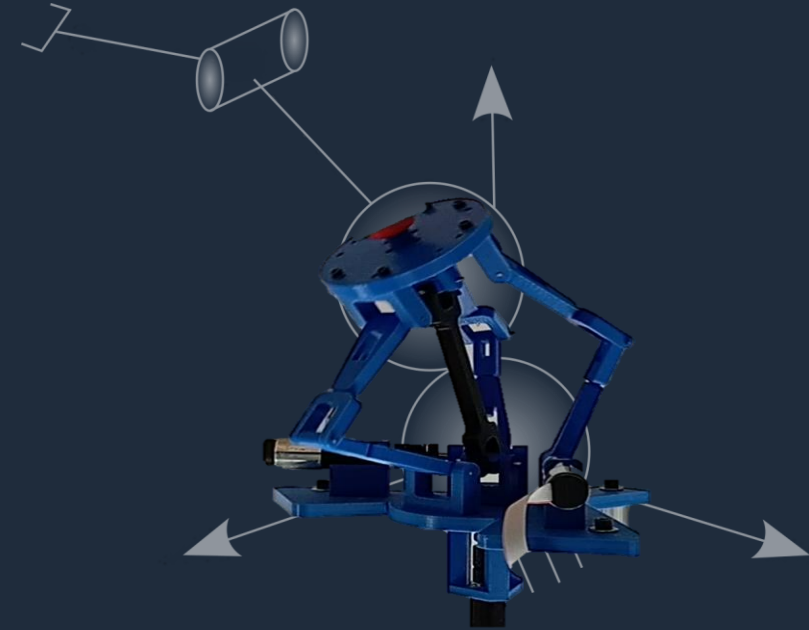
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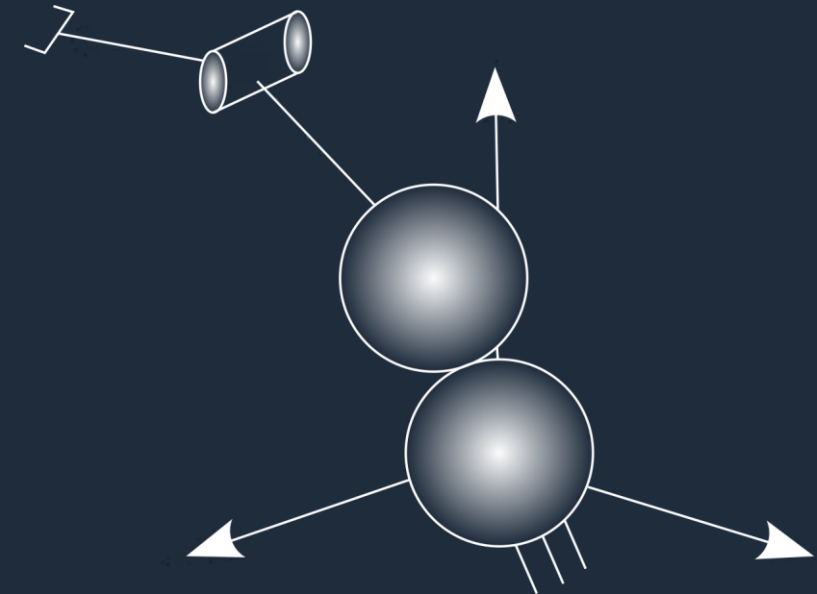
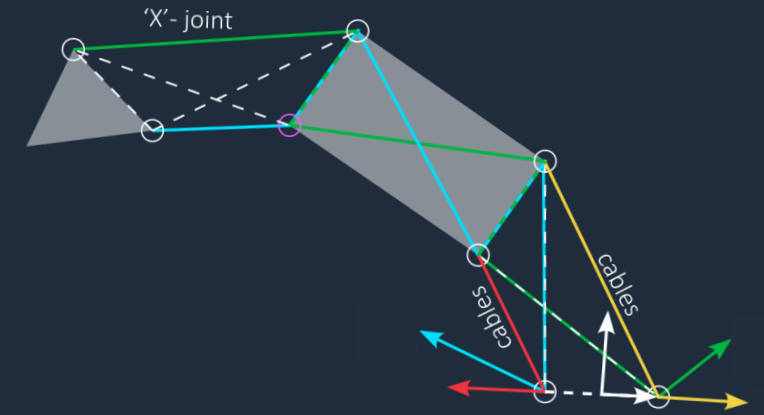


1. **Design:** Robots are either precise and rigid or soft and compliant.
2. **Control:** Explainable skill transfer framework that works across present robots and future designs missing.

2. Explainable skill transfer through Kinematic Intelligence



Older generation robots



Novel mechanisms

Phase 2

Mid-term scientific focus

Skill Scalability



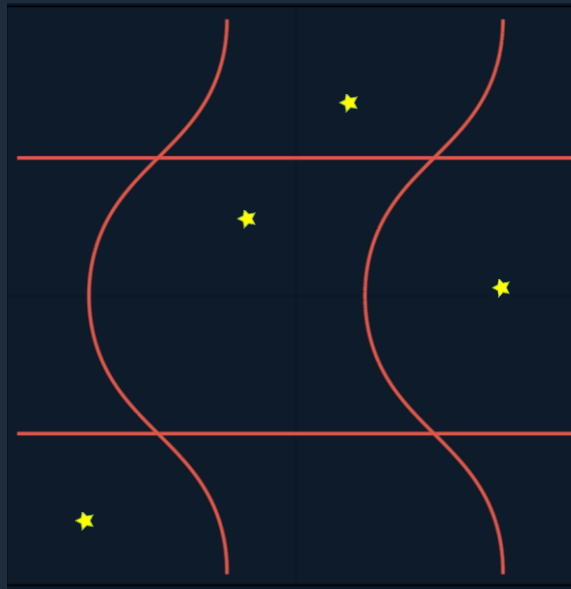
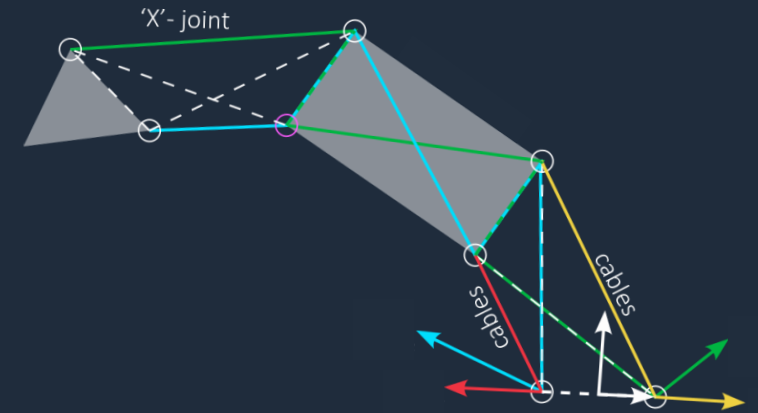
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Co-design approach:

1. **Design:** Rigid–soft trade-off through novel joints and mechanisms
2. **Control:** Explainable skill transfer through Kinematic Intelligence

2. Explainable skill transfer through Kinematic Intelligence



Example singularities and IK distribution that characterizes a robot

Integration

Choice of host laboratories



Team RDH

Le laboratoire des sciences de l'ingénieur, de l'informatique et de l'imagerie (UMR7357) (iCube)

Strasbourg



Team Gepetto

Laboratoire de recherche spécialisé dans l'analyse et l'architecture des systèmes (UPR 8001) (LAAS)

Toulouse

Parallel robots

Pierre Renaud

Medical robotics

Florent Nageotte
Benoit Rosa

Human-robot
interaction

Hassan Omran

Complements team's existing expertise

Kinematic
intelligence

Path Plannning

Design Optimisation

I bring my research expertise

Geometric Algebra

Olivier Piccin

Learning from
demonstration

Transfer learning

Nicolas Padoy

I contribute in expanding the current
expertise of the team

1. Collaborative robots, redundant robots (x 2) (Fig. A)
2. Haptic teleoperating system (Fig: B)
3. Strong collaboration with medical teams (IHU + Camma) (Fig. B, C)
4. Project **IMARA**: develop a lightweight robotic assistant for the operating room.



Figure A



Figure B



Figure C

1. Collaborative robots, redundant robots (x 2) (Fig. A)
2. Haptic teleoperating system (Fig: B)
3. Strong collaboration with medical teams (IHU + Camma) (Fig. B, C)
4. Project **IMARA**: develop a lightweight robotic assistant for the operating room.

Perfect environment to extend my research in medical applications



Figure A



Figure B



Figure C

Path planning

Optimisation

Transfer learning

Complements team's existing expertise

Philippe Souères
Florent Lamiroux

Nicolas Mansard

Olivier Stasse

Kinematic
Intelligence

Cuspidal robots

Parallel robots

I bring my research expertise

Co-design

Learning from
demonstration

I contribute in expanding the current
expertise of the team



Reliable Execution

Phase 1: Year 0-2

Cuspidal robots' singularities and control policy ambiguity

Use modern tools to study the workspace and control policy

Skill Scalability

Phase 2: Year 2-5

Skill transfer among current and future generations of robots

Kinematic Intelligence for skill transfer in current & future robots

Embodied Alignment

Phase 3: Year 5+

Learning based interactions need to be analytically grounded

Explore coupling of motion with language-based task planning

Co-design approach



Publications

- **8** journal articles: Science Robotics, IJRR, IEEE RA-L, MMT (x2), RCIM, JMD, SI
- **13** international conferences: ICRA, IROS, ARK, ISSAC, EUCOMES, CK, IMA (x2), RAHA, CASE, CFM, RAAD
- **2** open-source softwares: ParaOpt, Skill Transfer (Apr, '26)

Communications

- **6** Invited talks: SIMERO'23, RICAM'24, EPFL, Strasbourg, Toulouse, Bengaluru
- **3** thematic events: JJCR, Swiss Robotics Day (x2)
- **1** Media article: EPFL News (April, 2026)

Teaching

- **20 hrs CM**: Introduction to Robotics, Ecole Centrale de Nantes
- **20 hrs CM**: Kinematically grounded motion planning, Ecole Polytechnique Federale de Lausanne
- **96 hrs TP + TD**: Design of robots, Collaborative robotics, Optimal kinematic design of robots

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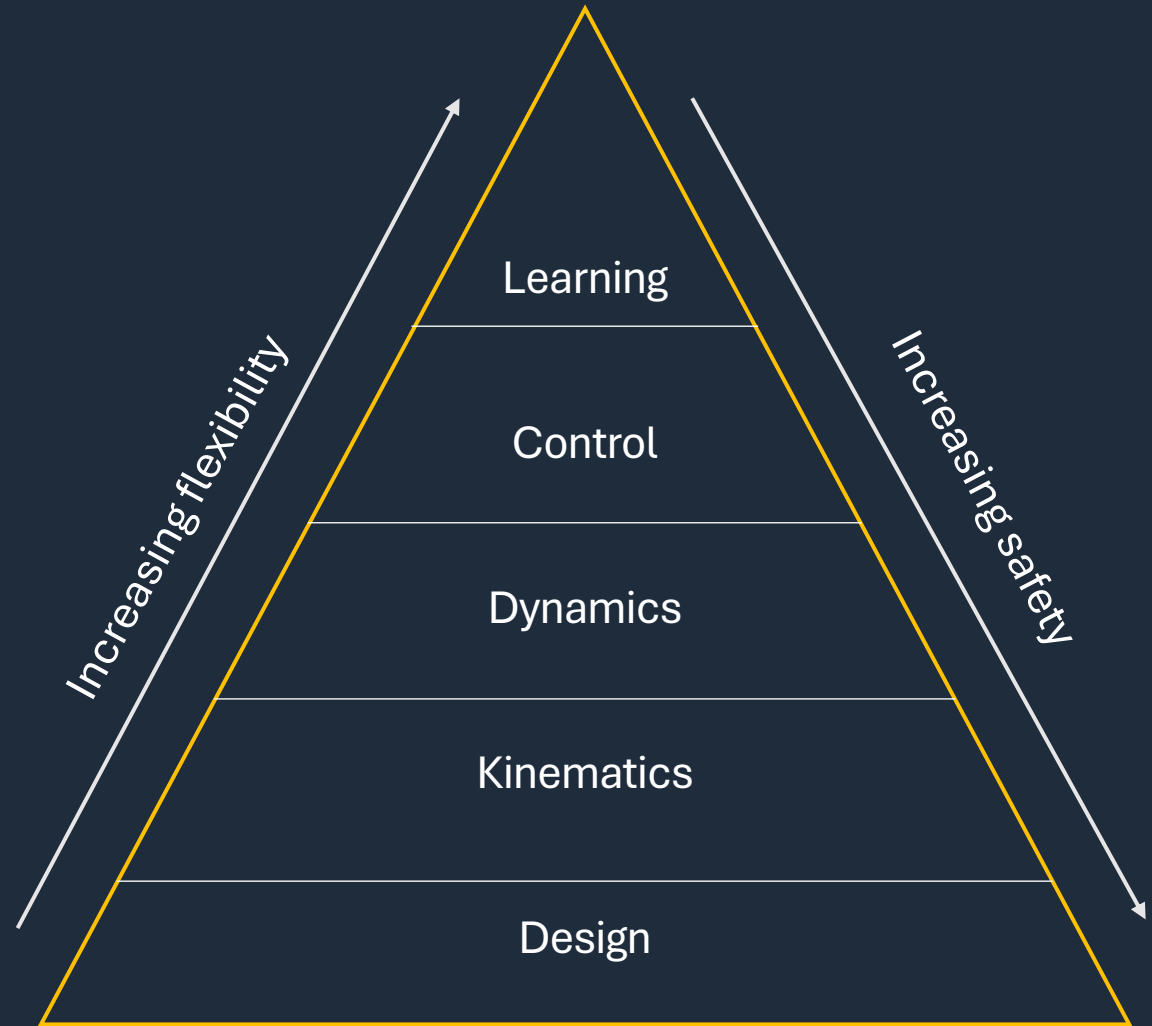
Phase 3

Long-term scientific focus

Grounded Interaction



1. Deterministic embedding of language with motion is essential
2. Designs developed using generative models need kinematic validation



Phase 3

Long-term scientific focus

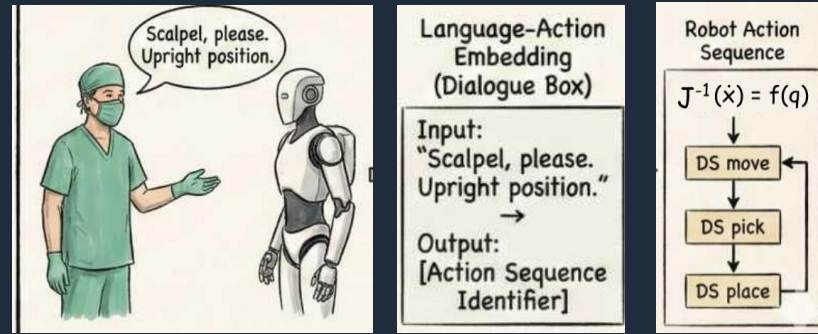
Grounded Interaction



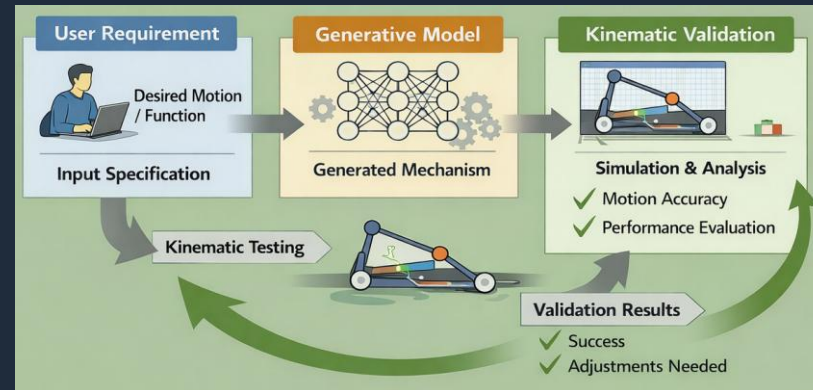
1. Deterministic embedding of language with motion is essential
2. Designs developed using generative models need kinematic validation

Co-design approach:

1. Develop motion language framework by mapping language tokens with DS
2. Develop generative models that are aware of kinematic analysis.



Language-based safe and explainable HRI



Kinematic validation of generative models

- Investigate the coupling of language and motion primitives
- Leverage task planning from language models, and guarantee motion with Kinematic intelligence
- Stage 1: Use a separate kinematic validation pipeline to verify designs from the generative model
- Stage 2: couple the kinematic understanding of the mechanisms in the model itself

Research activity

Doctoral work

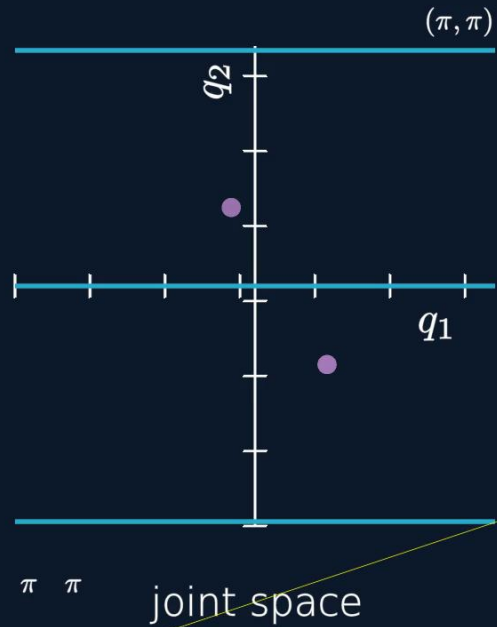


Path planning failure of cuspidal robots in real world application

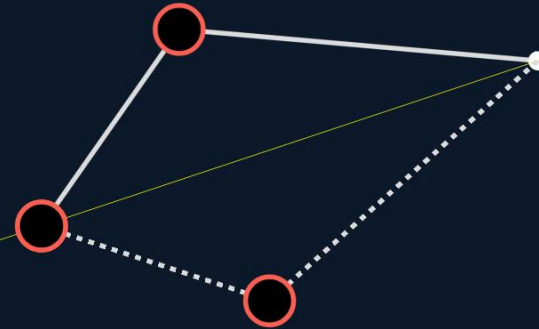
Source: Achille Verheye (achille0.medium.com)

Research activity

Doctoral work



Elbow Up configuration



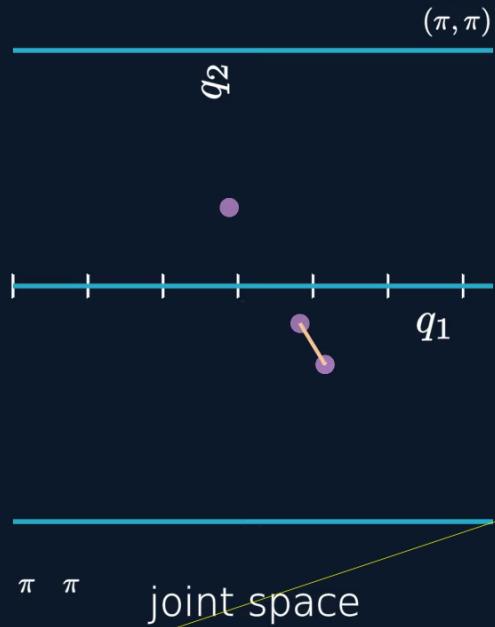
Elbow Down configuration

Noncuspidal robot

Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Research activity

Doctoral work



Elbow Up configuration

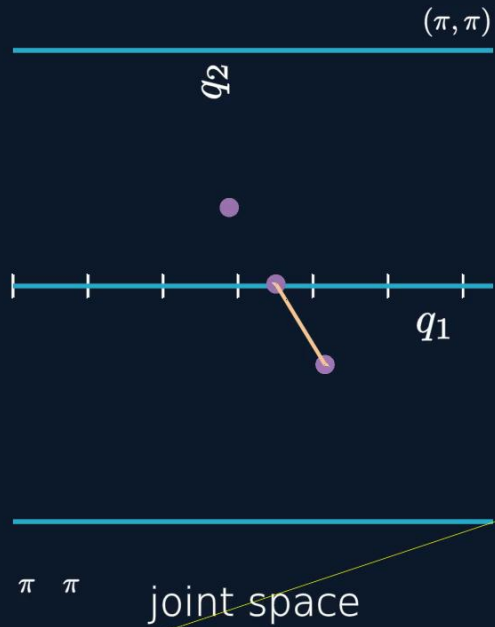
Elbow Down configuration

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Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Research activity

Doctoral work



Elbow Up configuration

A singular configuration is met while changing IKS

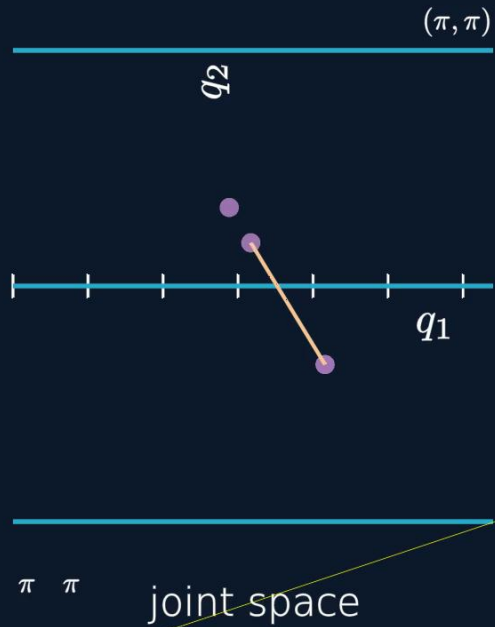
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Research activity

Doctoral work



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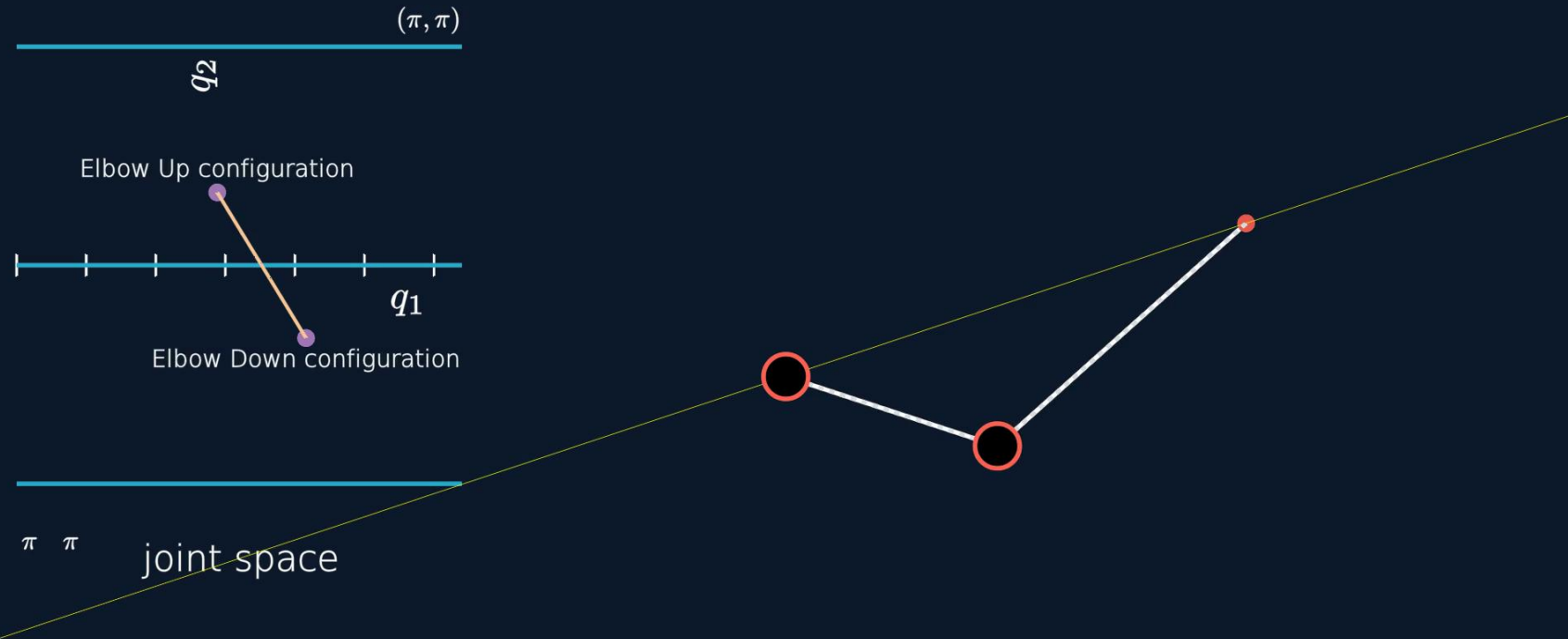
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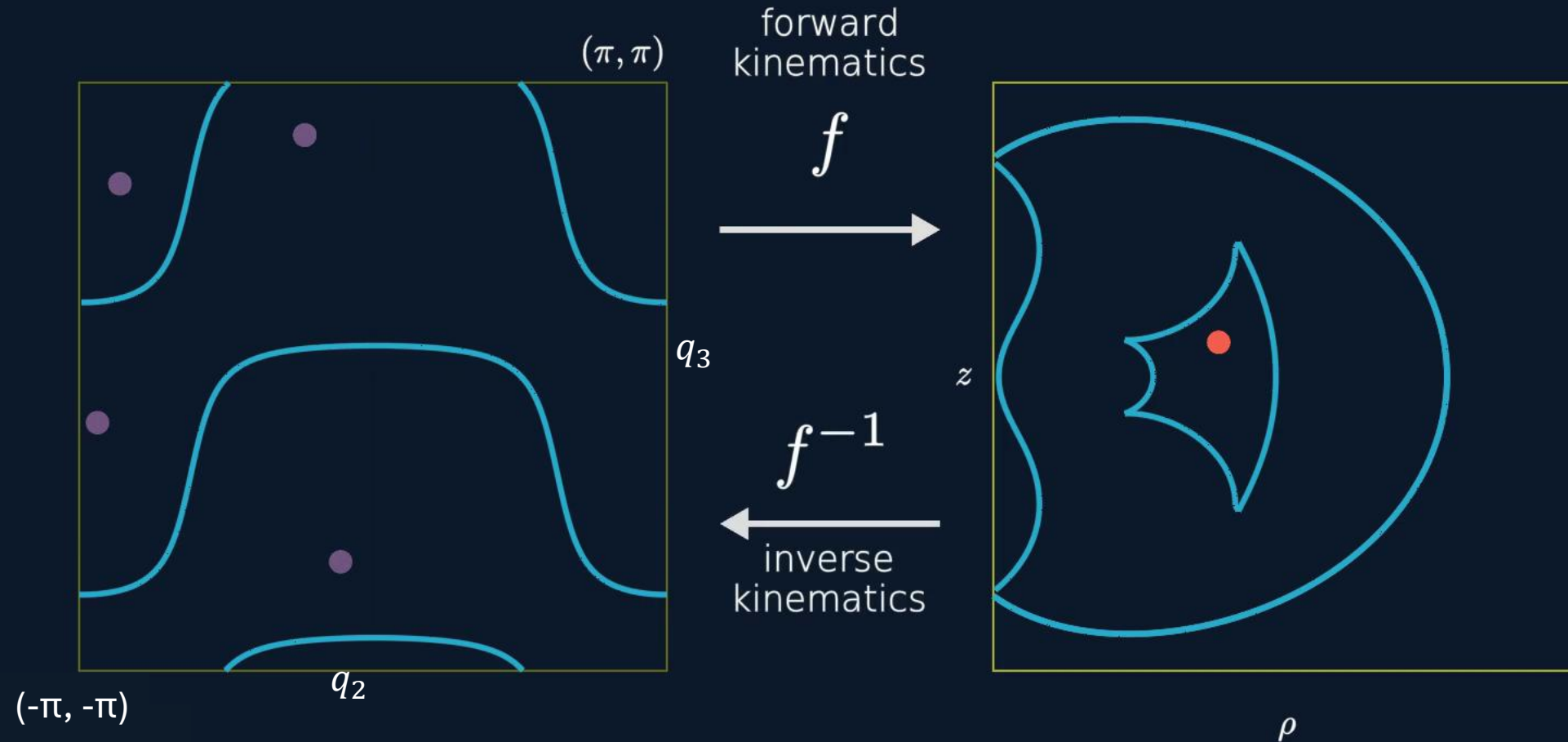


Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Noncuspidal robot

Research activity

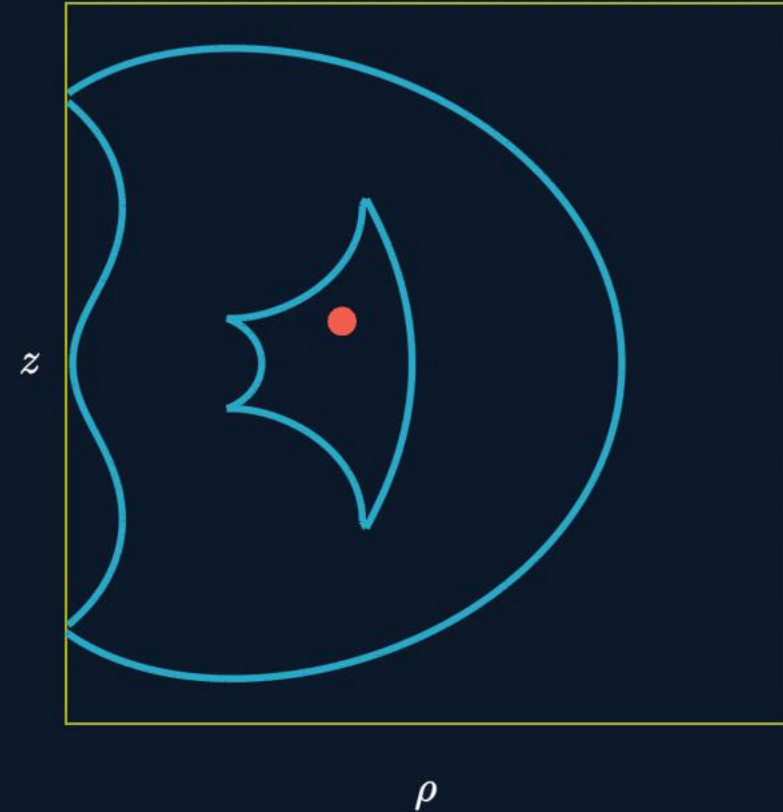
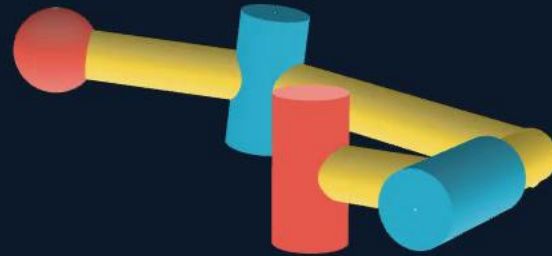
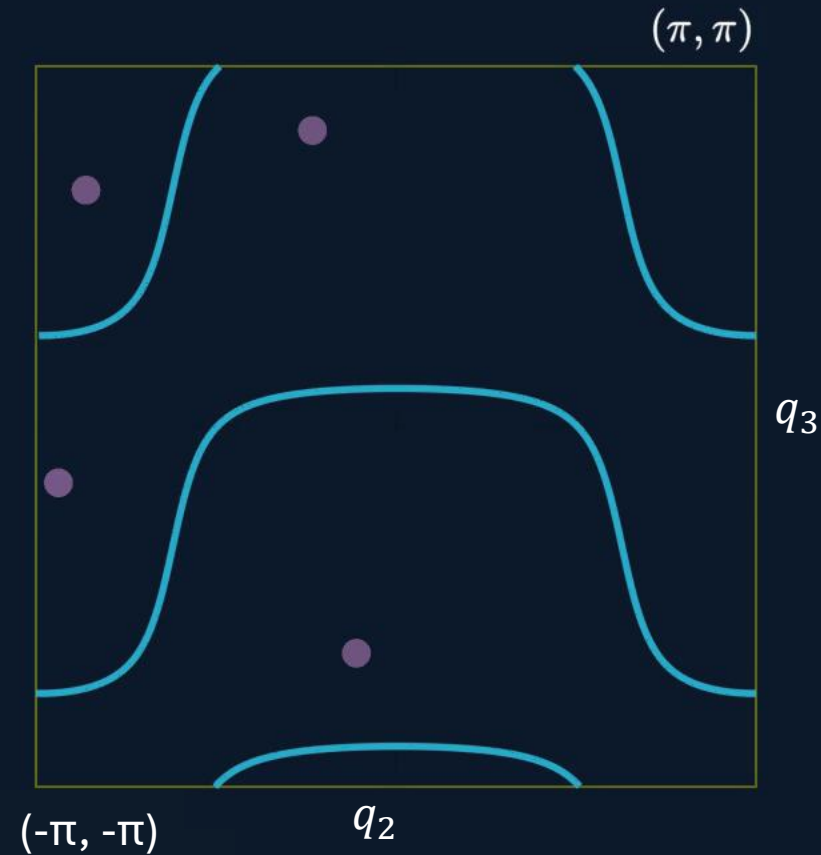
Doctoral work



Inverse kinematic solutions are **NOT** separated by singularities in cuspidal robots

Research activity

Doctoral work

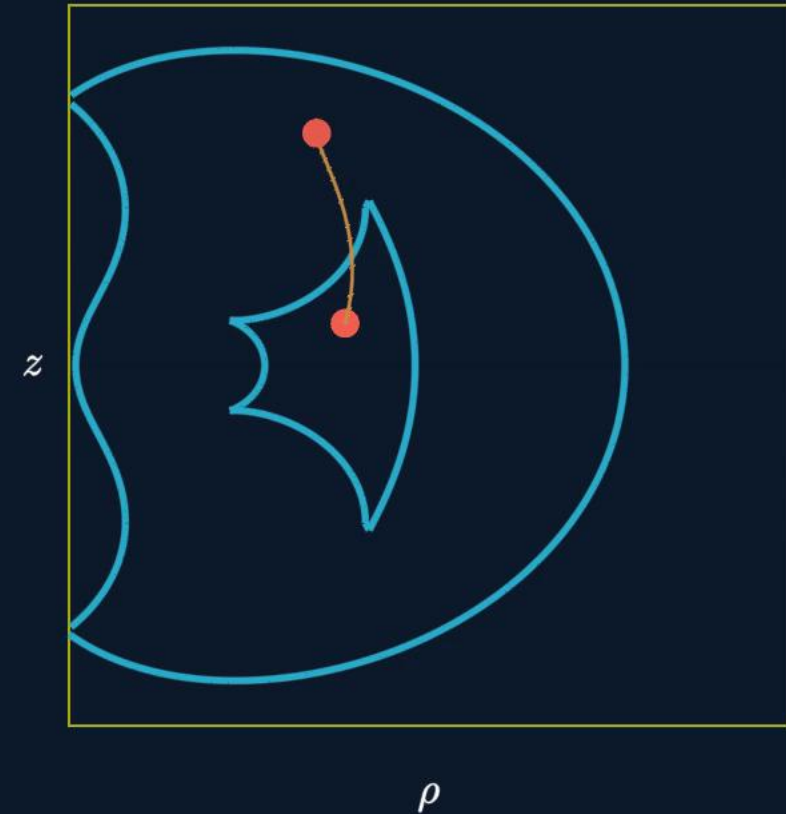
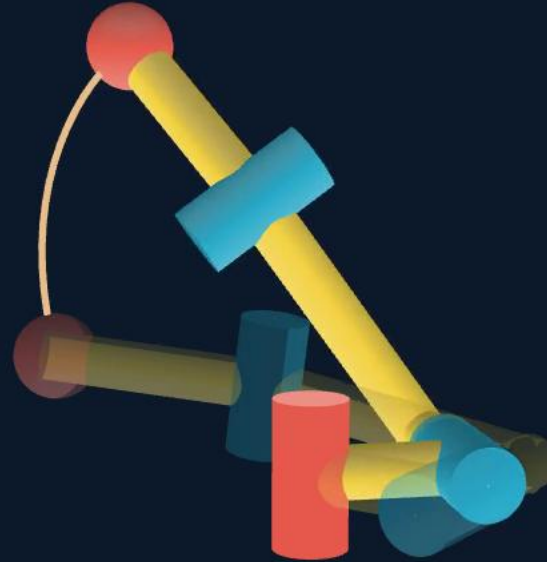
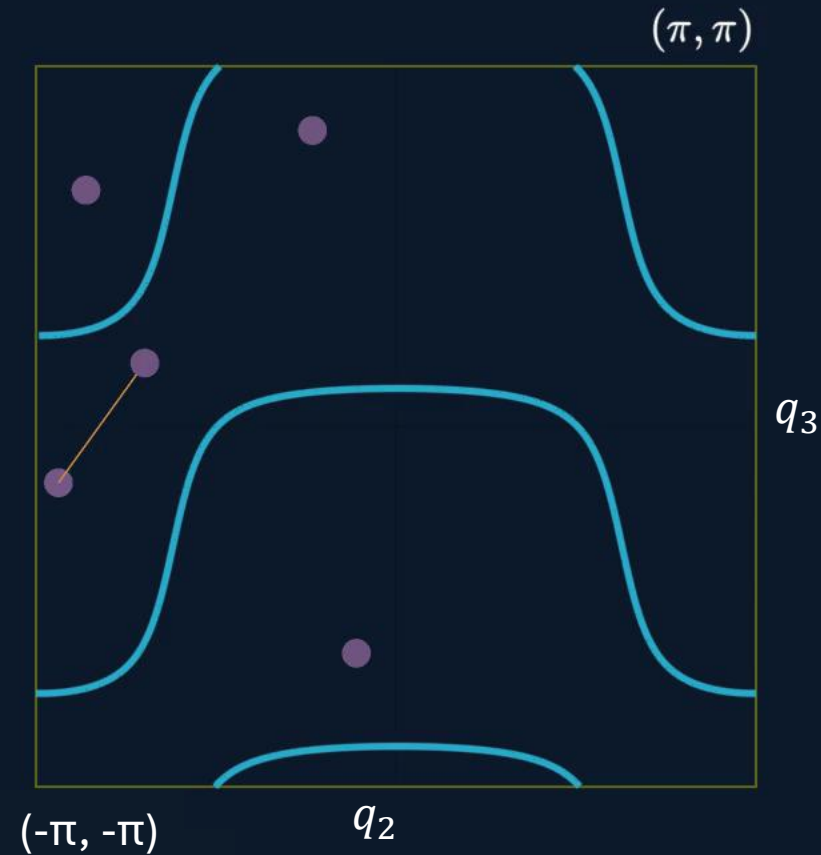


This leads to a change of solution without crossing singularities which causes issues such as **infeasible paths** and **non-repeatable paths**

$$\rho = \sqrt{x^2 + y^2}$$

Research activity

Doctoral work

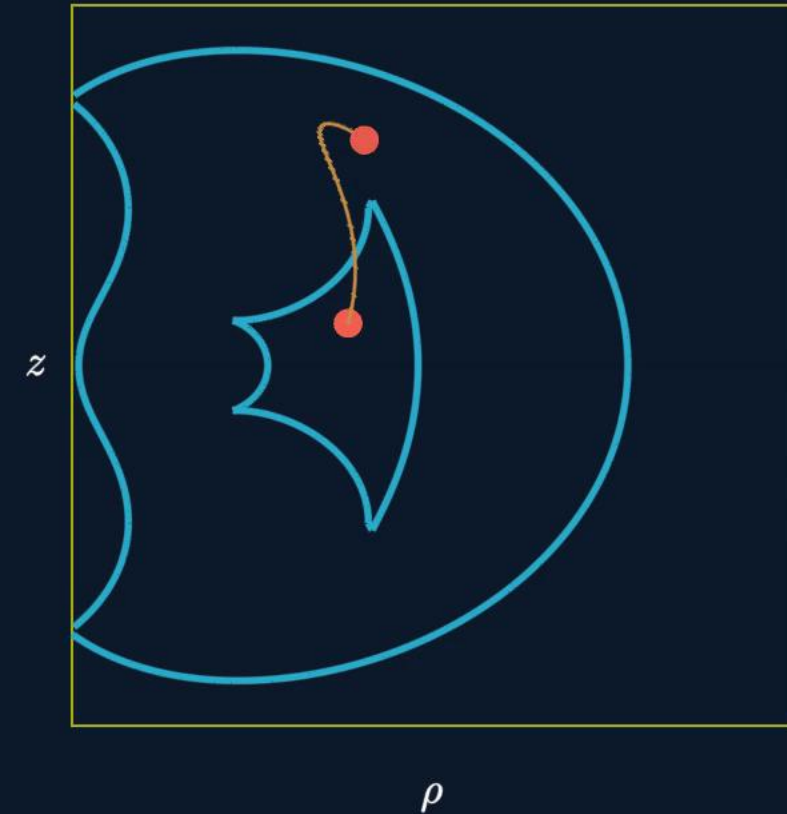
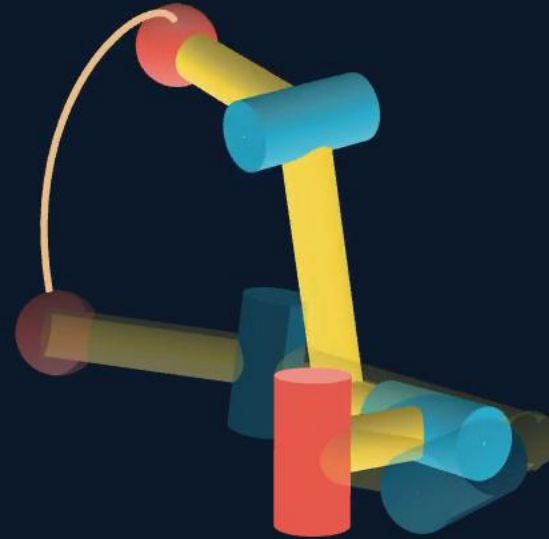
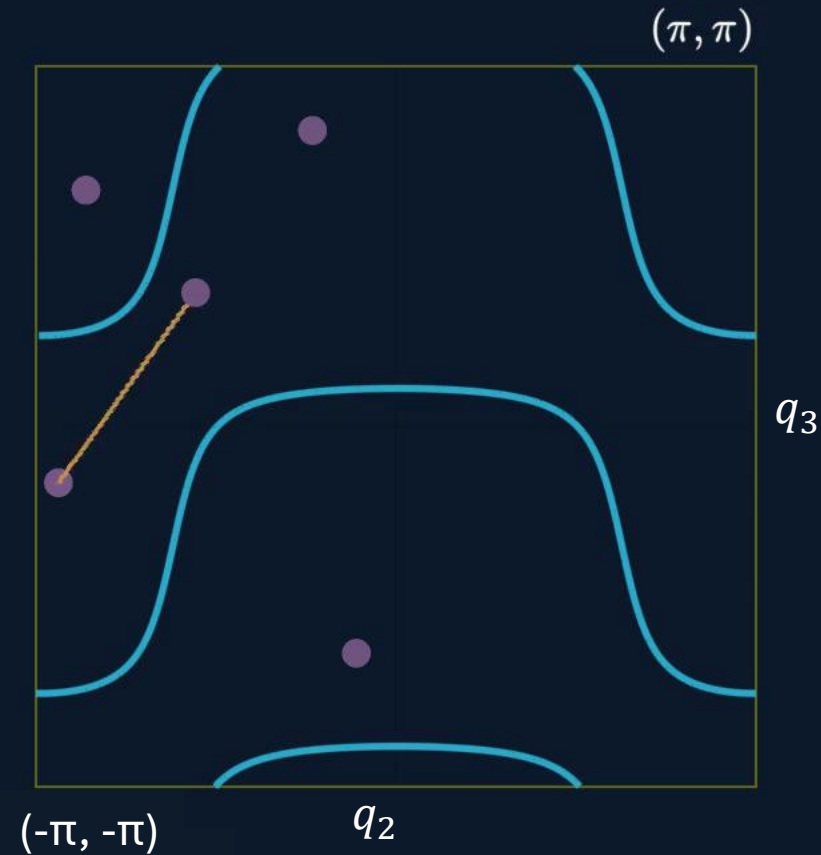


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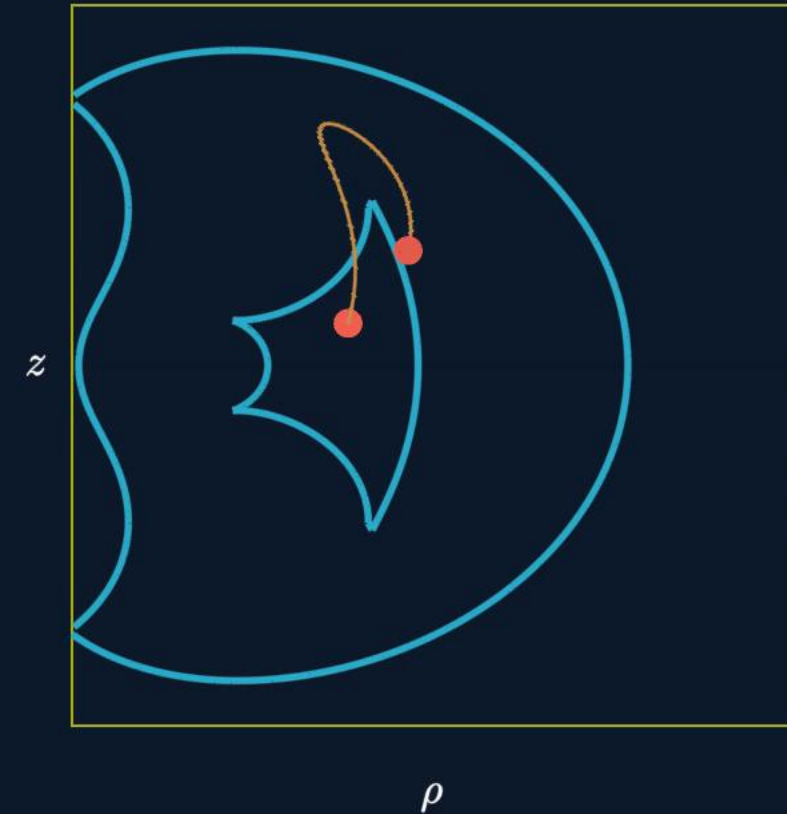
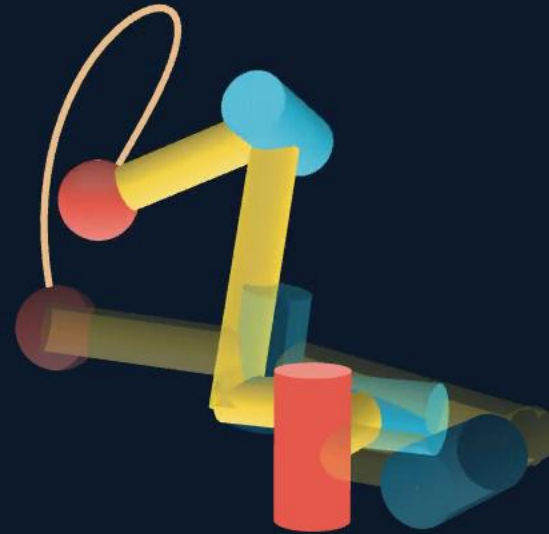
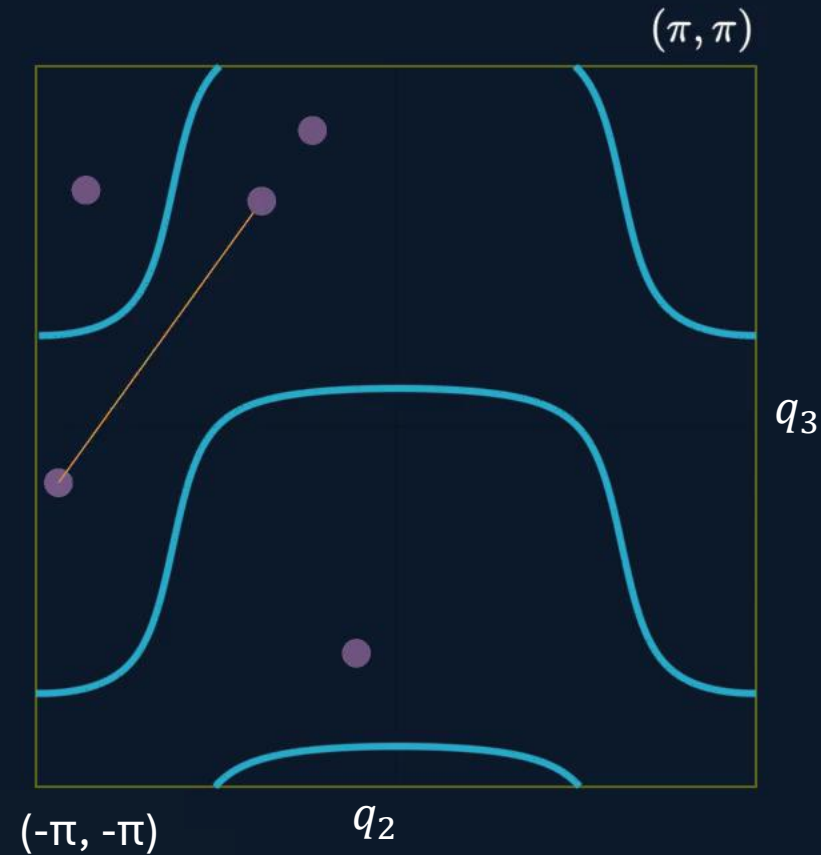


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Research activity

Doctoral work

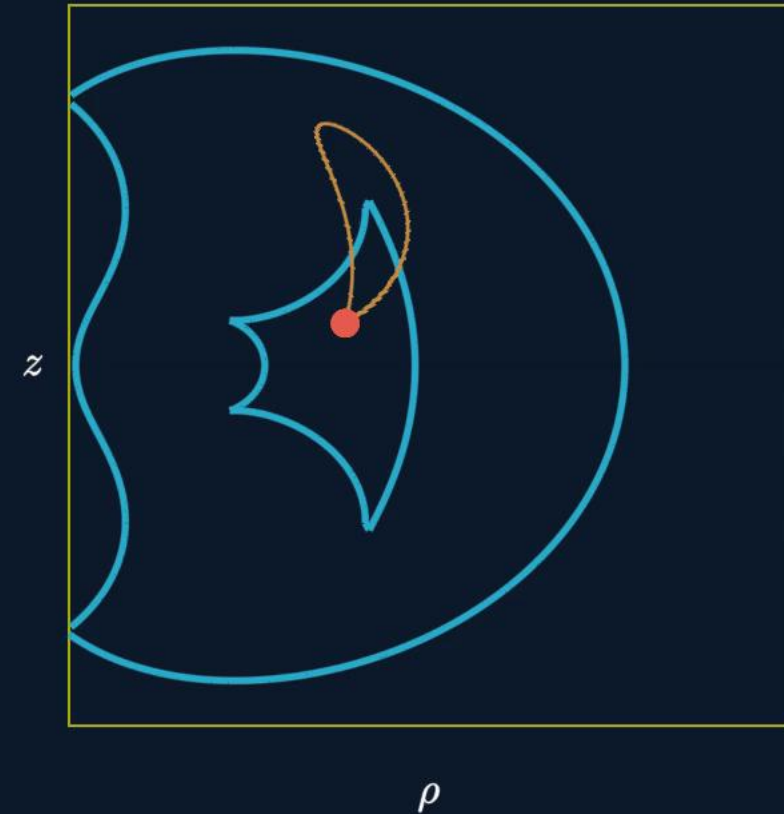
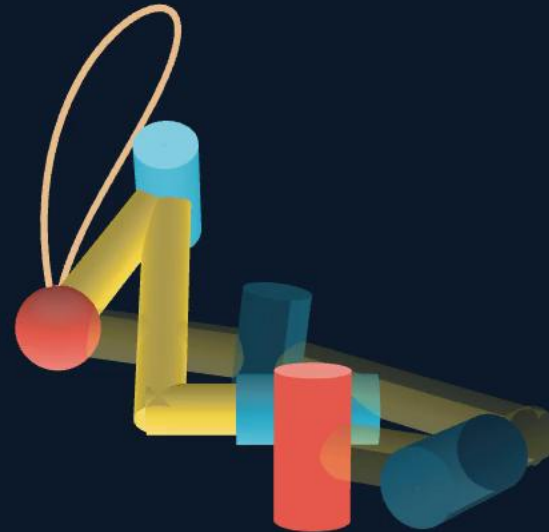
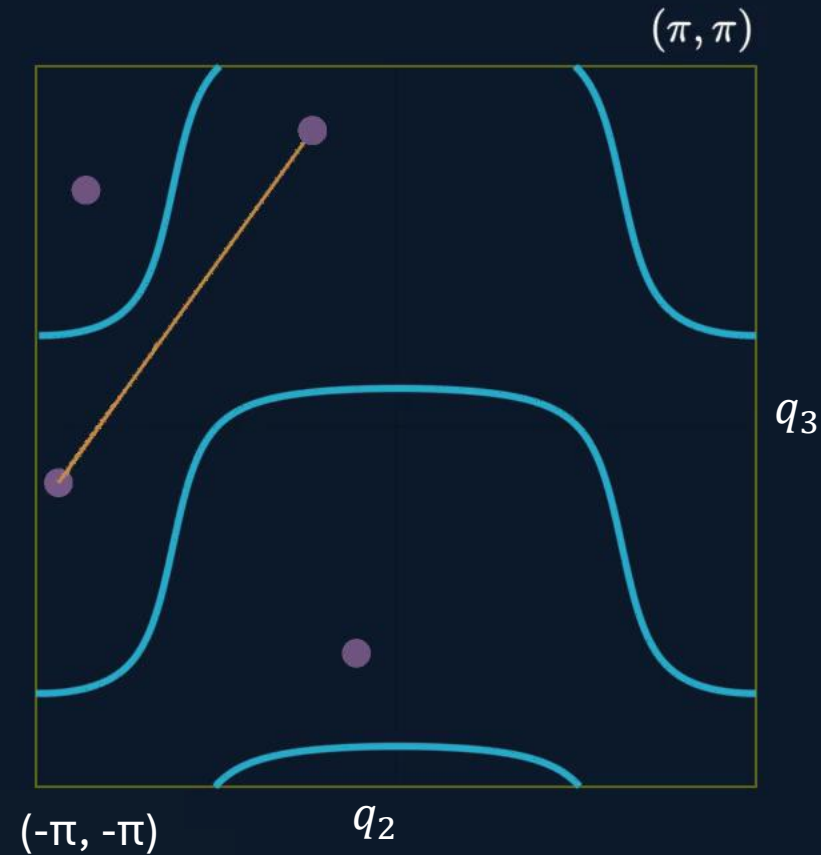


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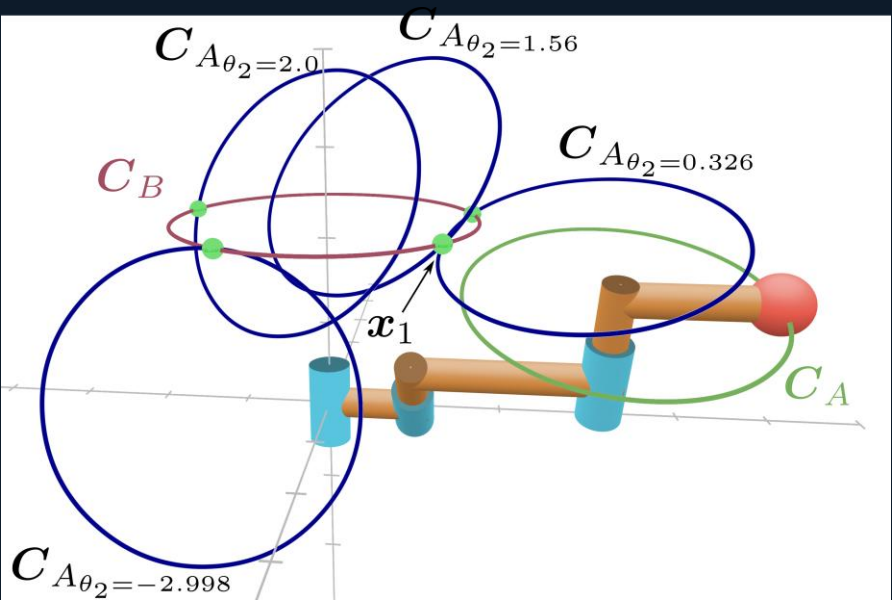


$$\rho = \sqrt{x^2 + y^2}$$

This leads to a change of solution without crossing singularities which causes issues such as **infeasible paths** and **non-repeatable paths**

What is Conformal Geometric Algebra?

- CGA is a framework to represent and manipulate:
 - Points, lines, planes, circles, spheres
 - Rigid & conformal transformations (rotate, translate, scale, invert)
- Embeds 3D space into 5D using:
- Two extra basis vectors: e_0 and e_∞
- Combines position + metric info
- Enables distance, angle, and intersection as algebraic operations



Inverse kinematics using CGA

$$X = \mathbf{x} + \frac{1}{2} ||\mathbf{x}||^2 e_\infty + e_0$$

Conformal embedding of a point

CGA primitives	Direct/primal (OPNS)	Dual (OPNS)
Point p	x	x
Point pair A	$p_1 \wedge p_2$	$S_1 \vee S_2 \vee S_3$
Sphere S	$p_1 \wedge p_2 \wedge p_3 \wedge p_4$	$p_S - \frac{1}{2}r^2 e_\infty$
Plane E	$p_1 \wedge p_2 \wedge p_3 \wedge e_\infty$	$n + d e_\infty$
Line L	$p_1 \wedge p_2 \wedge e_\infty$	$E_1 \vee E_2$
Circle C	$p_1 \wedge p_2 \wedge p_3$	$S_1 \vee S_2$ or $S_1 \vee E_1$

Motivation

Past work

PhD work

Postdoc work

Phase 1
Reliable execution

Phase 2
Skill Scalability

Phase 3
Embodied alignment

Integration

Cuspidal robots

C G A

Summary